# Supportive and Preventive Medicine

Sara Radparvar, Pharm.D., BCPS, BCCCP

The Mount Sinai Hospital

New York, New York

# Supportive and Preventive Medicine

Sara Radparvar, Pharm.D., BCPS, BCCCP

The Mount Sinai Hospital
New York, New York

**Learning Objectives**

1. Identify the key components of intensive care medicine that can be applied to all critically ill patients.
2. Recommend therapeutic options to prevent stress-related mucosal disease.
3. Recommend therapeutic options to prevent venous thromboembolism in a critically ill patient.
4. Compare therapeutic options for patients with heparin-induced thrombocytopenia.
5. Discuss medications that can be used to comfort a critically ill patient at the end of life.
6. Examine the pharmacists' role in disaster response and available resources.
7. Examine the role of social determinants of heath in patient outcomes.

**Abbreviations in This Chapter**

| | |
|---|---|
| aPTT | Activated partial thromboplastin time |
| CDI | *Clostridioides* (formerly *Clostridium*) *difficile* infection |
| DOAC | Direct oral anticoagulant |
| DVT | Deep venous thrombosis |
| $H_2RA$ | Histamine-2 receptor antagonist |
| HIT | Heparin-induced thrombocytopenia |
| ICU | Intensive care unit |
| NGT | Nasogastric tube |
| PF4 | Platelet factor 4 |
| PPI | Proton pump inhibitor |
| SRMD | Stress-related mucosal disease |
| SUP | Stress ulcer prophylaxis |
| VTE | Venous thromboembolism |

**Self-Assessment Questions**

*Answers and explanations to these questions may be found at the end of this chapter.*

1. On rounds, you have a "checklist" of interventions that will benefit all critically ill patients in an intensive care unit (ICU). Which interventions would be most effective?
   A. Initiate stress ulcer prophylaxis (SUP) in patients who are admitted to the ICU, and if appropriate, discontinue sedation.
   B. Initiate enteral nutrition, when appropriate, and mechanical venous thromboembolism (VTE) prophylaxis.
   C. If appropriate, initiate sedation interruption, and ensure that patients' heads are elevated 45 degrees above the bed, if not contraindicated.
   D. Assess the need for VTE prophylaxis in patients admitted to the ICU, and initiate an insulin infusion to maintain a blood glucose of 120 mg/dL.

2. Regarding pharmacologic prophylaxis for stress-related mucosal injury, which is most appropriate?
   A. Sucralfate neutralizes gastric pH.
   B. Proton pump inhibitors (PPIs) are superior to histamine-2 receptor antagonists ($H_2RAs$) in preventing clinically significant bleeding.
   C. Tolerance will occur with continued $H_2RA$ administration.
   D. Antacids are effective when used up to three times daily.

3. A 66-year-old man is admitted to the ICU with abdominal pain, nausea, and altered mental status. He has a history of alcoholic cirrhosis, atrial fibrillation, chronic kidney diease, and erosive esophagitis. He is intubated and stabilized on the ventilator. A nasogastric tube (NGT) is placed, and the patient is tolerating enteral tube feedings. Which would be best to recommend for preventing stress-related bleeding?
   A. Pantoprazole 40 mg intravenously twice daily.
   B. Ranitidine 50 mg intravenously three times daily.
   C. Famotidine 20 mg twice daily by NGT.
   D. Omeprazole suspension 20 mg once daily by NGT.

4. A 51-year-old woman is admitted to the ICU for hypovolemic shock secondary to severe dehydration. She reports a 5-day history of diarrhea and malaise. She has no recent history of illnesses or contact with health care personnel. Her medical history includes hypothyroidism and gastroesophageal reflux disease. Her medications include levothyroxine 25 mcg orally daily and famotidine 20 mg orally at bedtime. Recently, her primary care physician changed famotidine to omeprazole 20 mg orally at bedtime for increased gastroesophageal reflux disease symptoms. While she is in the ICU, testing for *Clostridioides* (formerly *Clostridium*) *difficile* infection (CDI) returns positive. Which would be most appropriate regarding PPI use and CDI?

A. PPIs are a potential risk factor for CDI by producing hypochlorhydria and increasing the host susceptibility to infections.
B. Prospective randomized controlled trials have shown that the risk of CDI is associated with PPI use.
C. There is no association between PPI use and CDI risk.
D. Studies reporting on CDI and PPI use have used the same definition of CDI and implemented the same infection control practices.

5. A 50-year-old woman (weight 70 kg) is admitted to the ICU for worsening mental status. Her medical history is significant for hypertension, tobacco use, and osteoporosis. The next morning, she is intubated and stabilized on a ventilator. An NGT is placed. Her current medications include ceftriaxone 2 g intravenously every 12 hours, vancomycin 1250 mg intravenously every 12 hours, acyclovir 800 mg intravenously every 8 hours, famotidine 20 mg by NGT twice daily, and a bowel regimen. Serum creatinine (SCr) is normal. Which would be the most appropriate VTE prophylaxis for this patient?

A. Intermittent pneumatic compression devices.
B. Enoxaparin 40 mg subcutaneously daily.
C. Unfractionated heparin continuous infusion to maintain a therapeutic activated partial thromboplastin time (aPTT).
D. No VTE prophylaxis at this time.

6. A 34-year-old woman (weight 65 kg) is admitted to the ICU with several upper extremity fractures, a closed-head injury, and a grade 4 liver laceration after a motor vehicle crash. Her medical history is nonsignificant. She is admitted to the ICU on a ventilator after surgery. Her current laboratory values are as follows: sodium (Na) 145 mEq/L, potassium (K) 3.1 mEq/L, chloride 97 mEq/L, carbon dioxide 18 mEq/L, blood urea nitrogen (BUN) 70 mg/dL, and SCr 3.5 mg/dL. Which would be the most appropriate VTE prophylaxis on the day of admission for this patient?

A. Provide intermittent pneumatic compression devices.
B. Give dalteparin 5000 units subcutaneously daily.
C. Give fondaparinux 2.5 mg subcutaneously daily.
D. No VTE prophylaxis is indicated at this time.

7. A 34-year-old man (weight 70 kg) is admitted to the surgical ICU for acute respiratory failure from pancreatitis. He has no pertinent medical history. His current medications include norepinephrine at 0.07 mcg/kg/minute, dexmedetomidine at 0.7 mcg/kg/hour, ampicillin/sulbactam 3 g intravenously every 6 hours, famotidine 20 mg intravenously twice daily, and heparin 5000 units subcutaneously three times daily. On day 3 of his ICU admission, the team suspects heparin-induced thrombocytopenia (HIT). His platelet count (Plt) was 360,000/$mm^3$ on admission and is 180,000/$mm^3$ today. The 4Ts score is used to determine the probability of HIT. The score is calculated as 3 equals low risk. The team would like to send the heparin–platelet factor 4 (PF4) immunoassay and initiate argatroban. Which is the most appropriate response?

A. Discontinue all heparin products, but do not initiate argatroban.
B. Discontinue all heparin products, and initiate argatroban.
C. Send the heparin-PF4 immunoassay, and continue low-dose unfractionated heparin until the results return.
D. Do not send the heparin-PF4 immunoassay, and continue low-dose unfractionated heparin.

8. Which would be the most important group of considerations in a critically ill patient approaching the end of life?

A. Pain management, tight glucose management, and control of secretions.
B. Routine vital sign checks, discontinuation of unnecessary medications, and control of secretions.
C. Pain management, control of secretions, and discontinuation of unnecessary medications.
D. Discontinuation of unnecessary medications, insertion of a Foley catheter, and treatment of nausea and vomiting.

***BPS Critical Care Pharmacy Examination Content Outline***

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
   a. Task A: 1–4
   b. Task B: 2, 5
2. Domain 2: Therapeutics and Patient Management
   a. Task A: 1–8
   b. Task B: 5, 9, 10, 13, 14
   c. Task C: 1, 3, 5
3. Domain 3: Professional Practice
   a. Task A: 2
   b. Task B: 1, 4, 7

## I. KEY ASPECTS IN THE GENERAL CARE OF ALL CRITICALLY ILL PATIENTS

A. The FAST-HUG mnemonic emphasizes important aspects of ICU medicine that can be applied at least daily to all critically ill patients to ensure safe, effective, and efficient care, as described in Table 1 (Crit Care Med 2005;33:1225-9).

**Table 1.** Key Elements of the FAST-HUG Approach

| Element | Importance | Considerations |
|---|---|---|
| **F**eeding | Malnutrition can lead to impaired immune function, in turn leading to increased susceptibility to infection, inadequate wound healing, bacterial overgrowth in the GI tract, and increased propensity for decubitus ulcers | • Initiate enteral feeding (preferred to parenteral nutrition) as soon as possible, typically within the first 24–48 hr after stabilization<br>• (See Fluids, Electrolytes, Acid-Based Disorders, and Nutrition Support chapter for a more in-depth review) |
| **A**nalgesia | Analgesic and sedative administration optimizes patient comfort and minimizes the acute stress response (hypermetabolism, increased oxygen consumption, hypercoagulability, and alterations in immune function) | • Pain should regularly be assessed with a validated tool such as the Behavioral Pain Scale (BPS) or the Critical-Care Pain Observation Tool (CPOT)<br>• Preemptive analgesia should be considered for invasive or potentially painful clinical procedures<br>• (See Management of Pain, Agitation, and Neuromuscular Blockade in Adult Intensive Care Unit Patients chapter for a more in-depth review) |
| **S**edation | | • Sedation should be assessed and reassessed with a validated tool such as the Richmond Agitation-Sedation Scale or the Sedation Agitation Scale<br>• Maintain light levels of sedation unless the patient has an indication for deep sedation, such as neuromuscular blockade<br>• If appropriate, execute daily sedative interruption<br>• (See Management of Pain, Agitation, and Neuromuscular Blockade in Adult Intensive Care Unit Patients chapter for a more in-depth review) |
| **T**hromboembolic prophylaxis | Most ICU patients have at least one risk factor for VTE | • Initiate appropriate prophylaxis, considering VTE and bleeding risks<br>• Mechanical prophylaxis (graduated compression stockings or intermittent pneumatic compression devices) is an alternative nonpharmacologic option in patients at high risk of bleeding |
| **H**ead of bed elevation | Elevating the head and thorax above bed to a 30–45 degree angle reduces the occurrence of GI reflux and nosocomial pneumonia in patients who are receiving mechanical ventilation | • Where appropriate, ensure patient position periodically throughout the day, especially after procedures that require the patient to lie flat |

**Table 1.** Key Elements of the FAST-HUG Approach *(continued)*

| Element | Importance | Considerations |
|---|---|---|
| Stress **U**lcer prophylaxis | Critically ill patients develop stress-related mucosal damage, potentially leading to clinically significant bleeding | • Consider discontinuing acid-suppressive medications when risk factors are no longer present |
| **G**lycemic control | Glycemic control is necessary in critically ill patients to decrease the incidence of complications such as decreased wound healing, increased infection risk, and increased risk of polyneuropathy | • Maintaining blood glucose at 140–180 mg/ dL should be considered in the acutely ill patient<br>• (See Fluids, Electrolytes, Acid-Based Disorders, and Nutrition Support chapter for a more in-depth review) |

GI = gastrointestinal; ICU = intensive care unit; VTE = venous thromboembolism.

B. Updated Mnemonic: FAST-HUGS BID (Crit Care Med 2009;37:2326-7; author reply 2327)
S = spontaneous breathing trial
B = bowel regimen
I = indwelling catheter removal
D = de-escalation of antimicrobials

C. MAIDENS: Additional Mnemonic with a Medication Focus (Can J Hosp Pharm 2011;64:366-9)
M = medication reconciliation
A = antibiotics or anti-infectives
I = indications for medications
D = drug dosing
E = electrolytes, hematology, and other laboratory results
N = no drug interactions, allergies, duplications, adverse effects
S = stop dates

D. Daily Checklists
1. Provide a framework of standardization and regulation of interventions in a systematic manner, allowing individuals to assess the presence or absence of the items.
2. Provide structure to important ICU-related interventions in an effort to reduce errors of omission and increase compliance with evidence-based practices to improve outcomes in the ICU patient population (N Engl J Med 2009;360:491-9; N Engl J Med 2006;355:2725-32)
3. Increase identification of drug-related issues (Can J Hosp Pharm 2013;66:157-62; Pharmacy [Basel] 2022 30;10:74)

**Patient Case**

1. A 68-year-old man (weight 85 kg) is admitted to the ICU for the management of severe hypoxemic respiratory failure associated with community-acquired pneumonia. He is endotracheally intubated and placed on mechanical ventilation. His medical history consists of Child-Pugh class B cirrhosis secondary to alcohol abuse, heart failure, and myocardial infarction. His laboratory values show white blood cell count (WBC) 15 x $10^3$ cells/$mm^3$, Plt 150,000/$mm^3$, BUN 15 mg/dL, SCr 1.1 mg/dL, K 4.5 mEq/L, international normalized ratio (INR) 1.0, aspartate aminotransferase (AST) 58 IU/mL, and alanine aminotransferase (ALT) 49 IU/mL. His current medications include azithromycin 500 mg intravenously daily, ceftriaxone 1 g intravenously daily, vancomycin 1250 mg intravenously every 12 hours, heparin 5000 units subcutaneously every 8 hours, fentanyl drip at 50 mcg/hour titrated to CPOT less than 2, midazolam drip at 1 mg/hour titrated to a Richmond Agitation-Sedation Scale (RASS) of 0 to -1, and a regular insulin drip at 1.5 units/hour titrated to maintain blood glucose at 140–180 mg/dL. Currently, on day 3 of his ICU stay, the patient's head is 30 degrees above the bed, his RASS is documented as -4, he is on minimal ventilator settings, and an NGT is placed. As the clinical pharmacist rounding on this patient, you go through the FAST-HUG mnemonic. Which are the best recommendations for the team?
   A. Initiate enteral nutrition by NGT, add SUP, and discontinue fentanyl and midazolam drips.
   B. Initiate enteral nutrition by NGT, discontinue deep venous thrombosis (DVT) prophylaxis, and transition the insulin drip to sliding scale.
   C. Transition the insulin drip to sliding scale, add SUP, and discontinue fentanyl and midazolam drips.
   D. Discontinue fentanyl and midazolam drips, discontinue DVT prophylaxis, and add SUP.

## II. STRESS ULCER PROPHYLAXIS

A. Epidemiology of Stress-Related Mucosal Disease (SRMD)
   1. Early studies demonstrated endoscopic evidence of superficial mucosal damage in 75%–100% of patients within 1–2 days after ICU admission.
      a. Few current evaluations are available
      b. Asymptomatic ulceration is likely clinically insignificant (Acta Anaesthesiol Scand 2017;61:216-23; Crit Care Med 1991;19:887-91)
   2. Incidence of clinically significant stress-related bleeding among the critically ill population is 2%–5% (Cureus 2020;12:e8029; Intensive Care Med 2019;45:1540-9)
   3. Stress ulcers and peptic ulcers are differentiated in Table 2

B. Characteristics of SRMD
   1. Several superficial erosive lesions occurring early in the course of critical illness, potentially progressing to deep ulcers
   2. Stress ulcers are diffuse and unamenable to endoscopic therapy; they generally heal over time, without intervention, as the patient's clinical status improves.

**Table 2.** Stress Ulcers vs. Peptic Ulcers

| Stress Ulcers | Peptic Ulcers |
|---|---|
| Several superficial lesions at the proximal stomach bulb; involves superficial capillaries; results from splanchnic hypoperfusion | Few deep lesions in the duodenum; typically involves a single vessel; results from break in gastric, duodenal, or esophageal lining from the corrosive action of pepsin |

C. Pathophysiology of SRMD (N Engl J Med 2018;378:2506-16)
   1. Decreased gastric blood flow and mucosal ischemia are the primary causes of stress ulcer–related bleeding.
   2. Reduced splanchnic blood flow is caused by mechanisms common to critical illness:
      a. Hypovolemia
      b. Reduced cardiac output
      c. Proinflammatory mediator release
      d. Increased catecholamine release
      e. Visceral vasoconstriction
   3. Additional factors leading to stress ulcer–related bleeding:
      a. Decreased gastric mucosal bicarbonate production
      b. Decreased gastric emptying of irritants and acidic contents
      c. Acid back-diffusion
      d. Reperfusion injury that may occur after restoration of blood flow after prolonged periods of hypoperfusion

D. Risk Factors for Stress-Related Bleeding
   1. Studies evaluating risk factors for stress-related bleeding in the presence and absence of SUP are summarized in Table 3.

**Table 3.** Studies Evaluating Risk Factors for Stress-Related Bleeding

| Study | Design and Population | Independent Risk Factors for Bleeding | Notes |
|---|---|---|---|
| N Engl J Med 1994;330:377-81 | • Multicenter, prospective cohort study of 2252 ICU patients<br>• Prophylaxis discouraged, except for head injury, burns > 30% of total body surface area, organ transplantation, evidence of peptic ulcer disease, or upper GI bleeding within previous 6 wk<br>• Prophylaxis administered for 674 patients (29.9%); agents included $H_2$RAs, antacids, sucralfate, prostaglandin analogues, and omeprazole | • Mechanical ventilation ≥ 48 hr<br>• Coagulopathy, defined as platelet count < 50,000 cells/mm$^3$, INR > 1.5, or activated partial thromboplastin time > 2 times control | • Patients with ≥ 2 risk factors had a 3.7% incidence of bleeding vs. 0.1% if risk factors were absent<br>• Most were cardiothoracic patients; extrapolations to other ICU settings potentially inaccurate |

**Table 3.** Studies Evaluating Risk Factors for Stress-Related Bleeding *(continued)*

| Study | Design and Population | Independent Risk Factors for Bleeding | Notes |
|---|---|---|---|
| Crit Care Med 1999;27:2812-7 | • Multicenter, prospective cohort study of 1200 mechanically ventilated patients<br>• Randomized trial of sucralfate vs. ranitidine for prevention of GI bleeding | • Renal failure (RR 1.16; 95% CI, 1.01–1.32). | • Enteral nutrition and SUP with ranitidine associated with lower incidence of GI bleeding |
| JAMA Intern Med 2014;174:564-74 | • Multicenter, retrospective database study of 35,312 ICU patients<br>• SUP administered with histamine-2 receptor antagonist or PPI for 48 hr | • Age > 50 yr<br>• Male sex<br>• Acute respiratory failure<br>• Shock<br>• Sepsis<br>• Acute kidney injury<br>• Acute or chronic hepatic injury<br>• Neurologic injury<br>• Myocardial infarction<br>• Coagulopathy | • Unable to assess severity of GI bleeding outcomes as events identified by ICD-9 codes |
| Intensive Care Med 2015;41:833-45 | • Multicenter, prospective, cohort study of 1034 ICU patients<br>• Overall, 73% received SUP at some point during ICU admission, most often PPIs | • Renal replacement therapy<br>• Liver disease<br>• Greater severity of illness | |

GI = gastrointestinal; ICU = intensive care unit; ICD-9 = *International Classification of Diseases, Ninth Revision*; INR = international normalized ratio; PPI = proton pump inhibitor; SUP = stress ulcer prophylaxis.

Information from: Cook DJ, Fuller HD, Guyatt GH, et al. Risk factors for gastrointestinal bleeding in critically ill patients. Canadian Critical Care Trials Group. N Engl J Med 1994;330:377-81; Cook DJ, Heyland D, Griffith L, et al. Risk factors for clinically important upper gastrointestinal bleeding in patients requiring mechanical ventilation. Canadian Critical Care Trials Group. Crit Care Med 1999;27:2812-7; MacLaren R, Reynolds PM, Allen RR. Histamine-2 receptor antagonists vs proton pump inhibitors on gastrointestinal tract hemorrhage and infectious complications in the intensive care unit. JAMA Intern Med 2014;174:564-74; Krag M, Perner A, Wetterslev J, et al. Prevalence and outcome of gastrointestinal bleeding and use of acid suppressants in acutely ill adult intensive care patients. Intensive Care Med 2015;41:833-45.

2. Additional risk factors for SRMD (Am J Health Syst Pharm 1999;56:347-79; J Trauma 1995;39:289-94):
   a. Spinal cord/head trauma
   b. Thermal injury affecting more than 35% of total body surface area
   c. History of GI bleed within the past year
   d. Postoperative transplantation
   e. Ulcerogenic medications (nonsteroidal anti-inflammatory drugs, aspirin, corticosteroids)
3. Enteral nutrition has been suggested as a protective variable (Crit Care Med 2010;38:2222-8; Crit Care 2018;43:108-13)

E. Pharmacologic Therapy for Preventing Stress Ulcers
   1. Antacids
      a. Dose-dependent neutralization of gastric acid
      b. Not recommended for routine use because of frequency of administration (up to every hour), adverse effects (diarrhea, constipation, electrolyte abnormalities), and interactions (interferes with the absorption of some drugs)

2. Sucralfate (Carafate)
    a. Complexes with albumin and fibrinogen to form a viscous, adhesive substance that adheres to ulcers with a gastric pH less than 4
    b. Not recommended for routine use because of adverse effects (constipation, aluminum toxicity, hypophosphatemia) and drug interactions by method of chelation; frequency of administration further complicates administration of other enteral medications
3. $H_2$RAs
    a. Available agents: Cimetidine (continuous infusion is the only $H_2$RA approved by the FDA for SUP), famotidine, nizatidine, and ranitidine
    b. Competitive blockade of histamine receptors on the basolateral membrane of the parietal cells. In addition, $H_2$RAs inhibit gastrin secretion to reduce acid production; however, they do not reliably inhibit vagally induced acid secretion.
    c. Alternative pathways (e.g., gastrin, acetylcholine) can be up-regulated in response to the use of $H_2$RAs, leading to a decrease in gastric pH and explaining the tolerance related to $H_2$RAs. Tolerance can occur as early as day 2 of $H_2$RA use (World J Gastrointest Pharmacol Ther 2014;5:57-62; Am J Gastroenterol 1999;94:351-7).
    d. In animal models, $H_2$RAs may also attenuate reperfusion injury by decreasing interleukin-6 and neutrophil activation, reducing inflammation by enhancing cell-mediated immunity, and acting as a weak free radical scavenger.
    e. Dose-dependent increase in gastric pH
    f. Previous studies of SRMD-related bleeding used either continuous infusion $H_2$RAs or combined $H_2$RAs with intermittent antacids to maintain a pH greater than 4. Current practice is to use intermittent administration of $H_2$RAs without pH monitoring.
    g. All $H_2$RAs require renal dosing adjustment.
    h. Adverse effects
        i. Mental status changes such as confusion, hallucinations, agitation, and headaches (mainly associated with cimetidine, but reported with famotidine and ranitidine in patients with severe renal failure)
        ii. Thrombocytopenia (occurs over several days from hapten formation; may occur within hours if patient is sensitized)
        iii. Rapid infusion–related hypotension
        iv. Prolongation of corrected QT interval has been reported with famotidine and ranitidine.
        v. Sinus bradycardia
        vi. Risk of nosocomial pneumonia
        vii. Increase in SCr with cimetidine – Competitively inhibits tubular secretion of creatinine
    i. Drug interactions
        i. Cimetidine inhibits cytochrome P450 (CYP) isoenzymes 3A4, 2D6, 2C9, 2C19, and 1A2.
        ii. pH-dependent interactions
    j. Available agents: Cimetidine (continuous infusion is the only $H_2$RA approved by the U.S. Food and Drug Administration [FDA] for SUP), famotidine, nizatidine, and ranitidine
4. PPIs
    a. Prodrugs activated in the acidic environment of the stimulated parietal cell inhibiting both histamine-induced and vagally mediated gastric acid by binding and inhibiting active proton pumps
    b. Dose-dependent increase in gastric pH, with maximal activity reached 3 days after initiation
    c. Most trials evaluated the effectiveness of enteral PPIs.
    d. Despite short elimination half-lives, PPIs suppress acid secretion for 20 hours or more, permitting once-daily dosing without requiring gastric pH monitoring.
    e. Tachyphylaxis does not occur with PPIs.

f. Rebound acid hypersecretion may occur after discontinuation; however, clinical relevance is unknown.
g. Adverse effects
    i. Diarrhea, abdominal pain, constipation, nausea
    ii. Headaches
    iii. Rash
    iv. Interstitial nephritis
    v. Hypomagnesemia (3 months or more of therapy)
    vi. Neurologic effects with high-dose intravenous omeprazole (hearing and vision disturbances)
    vii. Hypophosphatemia and metabolic alkalosis when administered with sodium bicarbonate
    viii. Vitamin $B_{12}$ deficiency
    ix. Increased risk of fractures (hip, waist, and spine)
    x. CDI (definitive cause-effect relationship is not well established)
    xi. Risk of nosocomial pneumonia
h. Drug interactions
    i. All agents are hepatically metabolized by CYP isoenzymes 3A4 and 2C19.
    ii. Omeprazole is an inhibitor of 3A4, 2C19, 2C9, and 1A2.
    iii. Lansoprazole may induce CYP1A2.
    iv. Clinically significant PPI interactions with clopidogrel through CYP2C19 inhibition have not consistently been shown to be important.
    v. pH-dependent interactions
i. Available agents: Dexlansoprazole, esomeprazole, lansoprazole, omeprazole (immediate-release capsule FDA approved for SUP), pantoprazole, and rabeprazole

F. Stress-Related Bleeding
1. Endoscopically evident mucosal damage and occult bleeding rates are reported from historical data; more contemporary data are lacking (Table 4).

**Table 4.** Categories of Stress-Related Bleeding

| Outcome | Incidence in ICU Patients | Definition |
|---|---|---|
| Endoscopically evident mucosal damage | 75%–100% | Superficial lesions identified on endoscopy |
| Occult bleeding | 15%–50% | Presence of guaiac-positive stools or nasogastric aspirate |
| Overt or clinically evident bleeding | 5%–25% | Appearance of coffee grounds in nasogastric aspirate, hematemesis, melena, or hematochezia |
| Clinically important bleeding | 1%–5% | Presence of overt bleeding with hemodynamic instability and/or blood transfusion within 24 hr of the event |

2. Clinically important GI bleeding
    a. Most trials define clinically important GI bleeding as overt bleeding accompanied by one of the following:
        i. Decrease in blood pressure of 20 mm Hg within 24 hours before or after GI bleeding episode
        ii. Decrease in blood pressure of 10 mm Hg and increase in heart rate of 20 beats/minute or more on orthostatic change.
        iii. Decrease in hemoglobin of 2 g/dL and transfusion of 2 units of blood within 24 hours of bleeding OR failure of the hemoglobin concentration to increase after transfusion by at least the number of units transfused minus 2 g/dL

b. Antacids, sucralfate, $H_2$RAs, and PPIs have all reduced clinically significant SRMD-related bleeding compared with placebo.

3. Studies evaluating interventions for stress-related bleeding are summarized in Table 5.

**Table 5.** Studies Evaluating Pharmacologic Interventions for Stress-Related Bleeding in Critically Ill Patients

| Comparison | Study | Design | Population | CIB | Notes |
|---|---|---|---|---|---|
| $H_2$RA vs Sucralfate | N Engl J Med 1998;338:791-7 | Multicenter, randomized, blinded, placebo-controlled | 1200 mechanically ventilated ICU patients | • CIB reduced with $H_2$RA (RR 0.44; 95% CI, 0.21–0.92) | • No difference in in mortality or VAP |
| | J Crit Care 2017;40:21-30 | Meta-analysis, 21 RCTs | 3121 ICU patients | • Similar risk of overt or CIB | • Conclusions limited by changes in practice (ventilator bundles, early nutrition) |
| $H_2$RA vs PPI | Crit Care 2016;20:120 | Meta-analysis, 19 RCTs | 2117 ICU patients | • CIB reduced with PPI (RR 0.39; 95% CI, 0.21–0.71; p=0.002) | • No difference in mortality or VAP<br>• Conclusions limited by disproportionate risk factors between patient groups, inconsistent definitions of bleeding, different routes/dosing of agents |
| | JAMA Intern Med 2014;174:564-74 | Retrospective cohort | 35,000 mechanically ventilated ICU patients | • CIB increased with PPI (OR 2.24; 95% CI, 1.81–2.76) | • Conclusions possibly limited by selection bias |
| | Chest 2018;154:557-66 | Retrospective cohort | 49,576 ICU patients | • CIB increased with PPI (HR 2.37; 95% CI, 1.61–3.5) | • Conclusions possibly limited by selection bias |
| | "PEPTIC" trial. JAMA 2020; 323:616-26 | Open-label, randomized, cluster crossover | 28,982 mechanically ventilated ICU patients | • CIB reduced with PPI (RR 0.73; 95% CI, 0.57–0.92)<br>• No difference in mortality (RR, 1.05; 95% CI, 1.00–1.10; p=0.54) | • Possible signal of harm with PPI use; increased mortality with PPI use in cardiac surgery population (RR 1.27; 95% CI, 0.51-1.09)<br>• Poor protocol adherence limits conclusions—only 63.4% in $H_2$RA group and 82.5% in PPI group exclusively received their assigned therapy |

**Table 5.** Studies Evaluating Pharmacologic Interventions for Stress-Related Bleeding in Critically Ill Patients (*continued*)

| Comparison | Study | Design | Population | CIB | Notes |
|---|---|---|---|---|---|
| PPI vs placebo | "SUP-ICU" trial. N Engl J Med 2018;379:2199-208 | RCTs | 3298 ICU patients | • CIB reduced with PPI (RR, 0.58; 95% CI, 0.40–0.86)<br>• No difference in mortality | • No difference in mortality, infectious complication or myocardial infarction<br>• Did not document enteral nutrition status |
| SUP vs placebo | Intensive Care Med 2020;46:1987-2000 | Meta-analysis, 74 RCTs | 39,569 ICU patients | • CIB likely reduced with PPI (RR 0.46; 95% CI 0.29–0.93) and $H_2$RA (RR 0.67; 0.48–0.94)<br>• Possible greater reduction with PPI | • No difference in VAP<br>• Potential for small increase in mortality with PPI |

CIB = clinically important bleeding; $H_2$RA = histamine-2 receptor antagonist; ICU = intensive care unit; PPI = proton pump inhibitor; RCT = randomized, controlled trial; SUP-stress ulcer prophylaxis; VAP-ventilator associated pneumonia.

Information from: Cook D, Guyatt G, Marshall J, et al. A comparison of sucralfate and ranitidine for the prevention of upper gastrointestinal bleeding in patients requiring mechanical ventilation. Canadian Critical Care Trials Group. N Engl J Med 1998;338:791-7; Alquraini M, Alshamsi F, Møller MH, et al. Sucralfate versus histamine 2 receptor antagonists for stress ulcer prophylaxis in adult critically ill patients: a meta-analysis and trial sequential analysis of randomized trials. J Crit Care 2017;40:21-30; Alshamsi F, Belley-Cote E, Cook D, et al. Efficacy and safety of proton pump inhibitors for stress ulcer prophylaxis in critically ill patients: a systematic review and meta-analysis of randomized trials. Crit Care 2016;20:120; MacLaren R, Reynolds PM, Allen RR. Histamine-2 receptor antagonists vs proton pump inhibitors on gastrointestinal tract hemorrhage and infectious complications in the intensive care unit. JAMA Intern Med 2014;174:564-74; Lilly CM, Aljawadi M, Badawi O, et al. Comparative effectiveness of proton pump inhibitors vs histamine type 2 receptor blockers for preventing clinically important gastrointestinal bleeding during intensive care: a population-based study. Chest 2018;154:557-66; PEPTIC Investigators for the Australian and New Zealand Intensive Care Society Clinical Trials Group, Alberta Health Services Critical Care Strategic Clinical Network, and the Irish Critical Care Trials Group; Young PJ, et al. Effect of stress ulcer prophylaxis with proton pump inhibitors vs histamine-2 receptor blockers on in-hospital mortality among ICU patients receiving invasive mechanical ventilation: the PEPTIC randomized clinical trial. JAMA 2020; 323:616-26; Krag M, Marker S, Perner A, et al. Pantoprazole in patients at risk for gastrointestinal bleeding in the ICU. N Engl J Med 2018;379:2199-208; Wang Y, Ge L, Ye Z, et al. Efficacy and safety of gastrointestinal bleeding prophylaxis in critically ill patients: an updated systematic review and network meta-analysis of randomized trials. Intensive Care Med 2020;46:1987-2000.

G. Infectious Complications
   1. Increases in gastric pH promote bacterial overgrowth, potentially leading to infectious complications.
   2. Both $H_2$RAs and PPIs cause changes in gastric pH; however, PPIs have a greater propensity to maintain a sustained higher pH.
   3. PPIs may also have immunosuppressive effects through the inhibition of neutrophils.
   4. Pneumonia
      a. Meta-analyses have shown lower pneumonia rates with sucralfate than with $H_2$RAs alone or $H_2$RAs combined with antacids (JAMA 1996;275:308-14; Crit Care Med 1991;19:942-9).
      b. Meta-analyses have failed to show an association between $H_2$RAs and PPIs on the risk of pneumonia (Crit Care Med 2013;41:693-705; Am J Gastroenterol 2012;107:507-20; Crit Care Med 2010;38:1197-205); however, a more recent meta-analysis showed an increased risk of pneumonia in patients receiving SUP and early enteral nutrition (Crit Care 2018;22:20).
      c. In two large pharmacoepidemiologic cohorts, an increase in the propensity-adjusted odds of pneumonia occurred with PPIs compared with $H_2$RAs (OR 1.2; CI, 1.03–1.41) in mechanically ventilated patients (JAMA Intern Med 2014;174:564-7) and in patients admitted for cardiac surgery after propensity matching (OR 1.19; CI, 1.03–1.38) (BMJ 2013;347:f5416).
      d. Many of the trials included in these analyses had varying definitions of pneumonia.

5. CDI
    a. A pharmacoepidemiologic cohort study found that CDI rates were significantly higher with PPIs than with $H_2$RAs (3.8% vs. 2.2%; p<0.001) (JAMA Intern Med 2014;174:564-7).
    b. A large retrospective study found PPI use to be an independent risk factor for developing CDI in medical ICU patients (OR 3.11; 95% CI, 1.11–8.74) (J Crit Care 2014;29:696).
    c. No prospective trials have been large enough to evaluate the risk of CDI in ICU patients.
    d. Many published trials have different definitions of CDI, unclear association of antisecretory therapy initiation and CDI diagnosis, and variable infection control practices.
6. In the SUP-ICU trial, there was no difference in infectious adverse events (new-onset pneumonia or CDI).
7. SUP duration should be evaluated daily, and SUP should be continued only as long as one or more risk factors are present.

H. Pharmacoeconomics
1. According to the landmark trial comparing $H_2$RAs with sucralfate, $H_2$RAs may be more cost-effective because of a reduced incidence of bleeding without an increase in pneumonia rates (N Engl J Med 1998;338:791-7).
2. Cost-effectiveness models have compared $H_2$RAs with PPIs with respect to clinically important bleeding and adverse effects (ventilator-associated pneumonia [VAP] and CDI).
    a. Use of PPI therapy for SUP resulted in a $1250 net cost savings per patient compared with $H_2$RAs. Univariate sensitivity analysis showed that PPI therapy was not as cost-effective when the probability of VAP rates was altered (Value Health 2013;16:14-22).
    b. Use of $H_2$RA therapy for SUP resulted in a $1095 net cost savings compared with PPIs. Univariate sensitivity analysis showed that assumptions of pneumonia and bleeding rates were the primary drivers of incremental costs (Crit Care Med 2014;42:809-15).
3. Initiating SUP in patients at risk and appropriately discontinuing SUP when a patient no longer has any of the risk factors for stress-related bleeding is the best practice for cost minimization.

I. Guideline Recommendations
1. In 1999, the first guideline from the American Society of Health-System Pharmacists was published (Am J Health Syst Pharm 1999;56:347-79). The guideline recommended that institutions decide on $H_2$RAs, antacids, or sucralfate according to safety profile, costs, and ease of administration.
2. In 2008, the Eastern Association for the Surgery of Trauma published a guideline recommending cytoprotective agents, $H_2$RAs, or PPIs; antacids were not recommended (www.east.org).
3. In 2014, the Danish Society of Intensive Care Medicine and the Danish Society of Anaesthesiology and Intensive Care Medicine published guidelines suggesting PPIs as the preferred agent (Dan Med J 2014;61:C4811).
4. In 2016, the Surviving Sepsis guidelines recommended PPIs or $H_2$RAs in patients at risk (Crit Care Med 2017;45:486-552).
5. The 2020 clinical practice guideline for gastrointestinal bleeding prophylaxis for critically ill patients recommends stress ulcer prophylaxis with a gastrointestinal bleeding risk of at least 4% based on several well-studied risk factors. The guidelines make a weak recommendation for PPIs over $H_2$RAs, and both PPIs and $H_2$RAs are strongly preferred over sucralfate. (BMJ 2020;368:l6722).

**Patient Case**

*Questions 2–4 pertain to the following case.*
A 45-year-old woman is admitted to the ICU for severe respiratory failure from community-acquired pneumonia. She is endotracheally intubated and placed on mechanical ventilation. An NGT is placed to begin enteral nutrition. Three days later, she is receiving norepinephrine and vasopressin infusions and appropriate antimicrobial agents. Her WBC is 20 x $10^3$ cells/$mm^3$, Plt 45,000/$mm^3$, BUN 70 mg/dL, SCr 4.5 mg/dL (baseline 0.9 mg/dL), K 4.5 mEq/L, INR 1.4, AST 30 IU/mL, and ALT 46 IU/mL.

2. Which best reflects this patient's risk factors for stress-related bleeding?
   A. Mechanical ventilation.
   B. Mechanical ventilation, coagulopathy, and acute kidney injury.
   C. Mechanical ventilation, coagulopathy, acute kidney injury, and shock.
   D. Mechanical ventilation, coagulopathy, acute kidney injury, shock, and enteral nutrition.

3. Which would be most appropriate for preventing stress-related bleeding?
   A. Sucralfate 1 g four times daily by NGT.
   B. Magnesium hydroxide 30 mL every 4 hours by NGT.
   C. Pantoprazole 40 mg intravenously twice daily.
   D. Famotidine 20 mg intravenously daily.

4. One week later, the patient's respiratory status has greatly improved. She has been off sedation and vasopressors for the past 4 days, working with physical therapy, and is now extubated. Her only medications include ceftriaxone, heparin subcutaneously, and SUP. Her current laboratory values are as follows: WBC 6 x $10^3$ cells/$mm^3$, Plt 256,000/$mm^3$, BUN 10 mg/dL, SCr 1.1 mg/dL, K 4.0 mEq/L, INR 0.8, AST 15 IU/mL, and ALT 10 IU/mL. Which would be the most appropriate recommendation regarding this patient's SUP regimen?
   A. SUP should be continued until hospital discharge.
   B. SUP should be continued until ICU discharge.
   C. SUP should be discontinued now.
   D. SUP should be discontinued once the patient is off antimicrobials.

## III. PROPHYLAXIS AGAINST DEEP VENOUS THROMBOSIS OR PULMONARY EMBOLISM

A. Epidemiology
   1. Reported occurrence of DVT is 10%–80% (J Crit Care 2002;17:95-104). Precise incidence in the critically ill population is challenging because of inconsistencies in patient populations, different diagnosis strategies, and variable study methodologies.
   2. DVT rates in the absence of prophylaxis vary, depending on the patient population.
      a. In the absence of prophylaxis: 30% in medical-surgical patients, 50%–60% in trauma patients, up to 80% in orthopedic surgical patients, and 20%–50% in neurosurgical patients (Arch Intern Med 2001;161:1268-79)

b. A randomized controlled trial of medical-surgical ICU patients receiving pharmacologic prophylaxis found proximal DVT rates of 5%–6% using ultrasonography twice weekly for detection (N Engl J Med 2011;364:1305-14).

B. Risk Factors
1. Malignancy, previous VTE, immobility, known thrombophilia, recent (1 month or less) surgery or trauma, older age (70 or older), heart or respiratory failure, sepsis, obesity (body mass index of 30 $kg/m^2$ or more), pregnancy, erythropoiesis-stimulating agents with a hemoglobin of 12 g/dL or more, hormonal therapy, recent transfusions of concentrated clotting factors, central venous lines, and long-distance travel (Chest 2012;141:S195-226)
2. Additional VTE risk factors in critically ill patients: A single-center prospective cohort (n=261) identified four independent risk factors for ICU-acquired VTE: personal or family history of VTE (multivariate hazard ratio [HR] 4.0; 95% CI, 1.5–10.3; p=0.004), end-stage renal failure (HR 3.7; 95% CI, 1.2–11.1; p=0.02), platelet transfusion (HR 3.2; 95% CI, 1.2–8.4; p=0.02), and vasopressor use (HR 2.8; 95% CI, 1.1–7.2; p=0.03) (Crit Care Med 2005;33:1565-71).
3. In the critically ill population, there are no validated risk assessment models to estimate the risk of VTE.

C. Prevention of VTE in the General Critically Ill Patient Population
1. Routine ultrasound screening is not recommended (Chest 2012;141:S195-226).
2. Prophylactic use of inferior vena cava filters is not recommended (Chest 2012;141:S195-226).
3. Mechanical VTE prophylaxis should be used in a critically ill patient if the patient is bleeding or at high risk of bleeding. Once the bleeding risk abates, pharmacologic VTE prophylaxis can be initiated.
   a. Intermittent pneumatic compression devices and graduated compression stockings (GCS) significantly reduce the risk of symptomatic VTE compared with no prophylaxis (Chest 2012;141:S195-226).
   b. A Cochrane review evaluated the results from 19 studies of GCS (Cochrane Database Syst Rev 2014;12:1-72).
      i. DVT rates were 9% in the GCS group and 21% in the control group (OR 0.33; p<0.00001), and pulmonary embolism rates were 2% in the GCS group and 5% in the control group (OR 0.38; p=0.04).
      ii. The patient population was largely limited to those undergoing orthopedic and general surgery.
   c. Data are limited in medical critically ill patients.
   d. The PREVENT trial randomized 2003 critically ill patients to receive either pharmacological prophylaxis alone or pharmacological prophylaxis with intermittent pneumatic compression devices. A majority of included patients were medical (>75%). The addition of mechanical prophylaxis did not result in a difference in the incidence of proximal lower-limb DVT (adjusted relative risk: 0.89; 95% CI, 0.58–1.36), pulmonary embolism (aRR 0.92; 95% CI, 0.32–2.06), and the composite of lower limb DVT/pulmonary embolism (aRR 1.11; 95% CI, 0.85–1.44) (N Engl J Med 2019;380:1305-15).
4. Either low-dose unfractionated heparin or low-molecular-weight heparin should be initiated in a critically ill patient over no prophylaxis.
5. The 2018 American Society of Hematology guidelines recommend continuing VTE prophylaxis only while patients remain admitted in the hospital, over continuing extended prophylaxis (Blood Adv 2018;2:3360-92).
6. Randomized trials of VTE prophylaxis in critically Ill patients are summarized in Table 6.
7. Recommendations for VTE prophylaxis in trauma patients are summarized in Table 7.

8. Recommendations for VTE prophylaxis in the general and abdominal-pelvic surgical patient are summarized in Table 8.
9. Anticoagulants commonly employed and their usual doses for VTE prophylaxis are summarized in Table 9.

**Table 6.** Randomized Trials of VTE Prophylaxis in Critically Ill Patients

| Citation | Study Type | Population | Intervention | Screening Methods | VTE Rates | Major Bleeding Rates |
|---|---|---|---|---|---|---|
| Crit Care Med 1982;10:448-50 | Single-center | 119 medical-surgical ICU patients | LDUH 5000 units SC twice daily vs. placebo | Daily 125I-labeled fibrinogen leg scanning | DVT: 13% in LDUH group vs. 29% in placebo group; p<0.05 | NR |
| N Engl J Med 1996;335:701-7 | Single-center | 344 patients with major trauma | Enoxaparin 30 mg twice daily vs. LDUH 5000 units twice daily | Venography on days 10–14 and US if VTE suspected | Proximal DVT: 14.7% in LDUH group vs. 6.2% in enoxaparin group; p=0.012<br>PE: 0% in LDUH group vs. 0.8% in enoxaparin group; p=NR | 0.6% in LDUH group vs. 2.9% in enoxaparin group; p=0.12 |
| Am J Respir Crit Care Med 2000;161:1109-14 | Multicenter, double-blind | 221 MV patients with COPD (169 patients evaluated) | Nadroparin (weight based) SC daily vs. placebo | Weekly US and venography | DVT: 15.5% in nadroparin group vs. 28.2% in placebo group; p=0.045 | 5.6% in nadroparin group and 2.7% in placebo group; p=0.28 |
| Thromb Haemost 2009;101:139-44 | Multicenter, double-blind | 1935 patients with severe sepsis receiving drotrecogin alfa (activated) | LDUH 5000 units SC twice daily vs. enoxaparin 40 mg SC daily vs. placebo | US on day 4–6 | DVT: 5.6% in LDUH group vs. 5.9% in enoxaparin group vs. 7.0% in placebo group; p=NS<br>PE: 0.4% in LDUH group vs. 0.4% in enoxaparin group vs. 0.8% in placebo group; p=NS | NR |
| Blood Coagul Fibrinolysis 2010;21:57-61 | Single-center, double-blind | 156 surgical patients undergoing major elective surgery | LDUH 5000 units SC twice daily vs. enoxaparin 40 mg SC daily | US 5–7 days after surgery and when clinically indicated | DVT: 2.7% in LDUH group vs. 1.2% in enoxaparin group; p=0.51 | 2.7% in LDUH group vs. 1.2% in enoxaparin group; p=0.48 |

**Table 6.** Randomized Trials of VTE Prophylaxis in Critically Ill Patients (*continued*)

| Citation | Study Type | Population | Intervention | Screening Methods | VTE Rates | Major Bleeding Rates |
|---|---|---|---|---|---|---|
| N Engl J Med 2011;364:1305-14 | Multicenter, double-blind | 3746 medical-surgical ICU patients expected to remain in the ICU ≥ 3 days (90% medical, 76% MV) | LDUH 5000 units SC twice daily vs. dalteparin 5000 international units SC daily | US 2 days after admission, twice weekly, and as clinically indicated | Proximal DVT: 5.8% in LDUH group vs. 5.1% in dalteparin group; p=0.57<br>PE: 2.3% in LDUH group vs. 1.3% in dalteparin group; p=0.01 | 5.6% in LDUH group vs. 5.5% in daltepa-rin group; p=0.98 |

COPD = chronic obstructive pulmonary disease; DVT = deep venous thrombosis; LDUH = low-dose unfractionated heparin; MV = mechanically ventilated; PE = pulmonary embolism; NR = not reported; NS = not significant; SC = subcutaneously; US = ultrasonography.

D. Prevention of VTE in the Non-orthopedic Surgical Patient (Chest 2012;141:S227-77)

**Table 7.** VTE Prophylaxis Recommendations in Trauma Patients

| Risk Level for VTE | Risk of Bleeding | Prophylaxis |
|---|---|---|
| Low-moderate | Low | LMWH,[a] LDUH,[a] or IPCD (all preferred to no prophylaxis) |
| High[b] | Low | LMWH[a] or LDUH[a] with elastic stockings or IPCD<br>The American Association for the Surgery of Trauma/ American College of Surgeons recommend VTE prophylaxis with enoxaparin, with an empiric dose of 40 mg SC every 12 hr adjusted based on anti-Xa level. Lower doses of 30 mg SC every 12 hr may be considered in patients > 65 years of age, < 50 kg, CrCl 30–60 ml/min or other risk factors for bleeding or drug accumulation (J Trauma Acute Care Surg 2022;92:597-604) |

[a]If LDUH or LMWH is contraindicated, mechanical prophylaxis with IPCD is preferred to no prophylaxis in the absence of lower-extremity injury.

[b]Includes acute spinal cord injury, traumatic brain injury, and spinal surgery from trauma; pharmacologic prophylaxis should be initiated as soon as possible, typically 24–48 hr after the event, but this may depend on the extent of bleeding on head computed tomography.

IPCD = intermittent pneumatic compression device; LMWH = low-molecular-weight heparin.

**Table 8.** VTE Prophylaxis Recommendations in the General and Abdominal-Pelvic Surgical Patient

| Risk Level for VTE | Risk of Bleeding | Prophylaxis |
|---|---|---|
| Very low | Low | Early ambulation |
| Low | Low | IPCD |
| Moderate | Low | LMWH, LDUH, or IPCD |
| | High | IPCD |
| High | Low | LMWH or LDUH with elastic stockings or IPCD |
| | Low with contraindications to LMWH or LDUH | Low-dose aspirin, fondaparinux, or IPCD |
| | High | IPCD until risk of bleeding abates; then pharmacologic prophylaxis should be initiated |

**Table 9.** Available Agents and Dosing

| Agent | Dose in Patients with Normal Renal Function | Dose in Patients with Renal Impairment[a] |
|---|---|---|
| Enoxaparin | 40 mg SC daily | 30 mg SC daily |
| Dalteparin | 5000 units SC daily | Specific dosage adjustments have not been recommended; accumulation did not occur in critically ill patients with severe renal insufficiency. (Arch Intern Med 2008;168:1805-12) |
| LDUH | 5000 units SC every 8–12 hr: Choosing between every 8 hr and every 12 hr should be based on the patient's risk of thrombosis and bleeding | |
| Fondaparinux | 2.5 mg SC once daily for patients weighing ≥ 50 kg | Contraindicated; however, doses of 2.5 mg SC every 48 hr have been used (Pharmacotherapy 2017;37:1241-48). |

[a]Estimated CrCl 20–30 mL/min/1.73 $m^2$.

CrCl = creatinine clearance.

E. Considerations for Critically Ill Patients
   1. Inability to communicate symptoms (impaired consciousness) and altered physical examination (edema) make the diagnosis of symptomatic VTE challenging in the critically ill population. Routine screening for VTE with ultrasonography is not recommended.
   2. Dosing frequency of low-dose unfractionated heparin (twice vs. thrice daily): Meta-analyses suggest no difference in thrombosis or major bleeding rates with twice-daily regimens compared with thrice-daily regimens in nonsurgical hospitalized patients (Chest 2011;140:374-81). Most of these patients were not critically ill and had few comorbidities. A more recent observational study using the Premier Healthcare Database compared thrice-daily with twice-daily dosing in 30,800 critically ill patients. No significant differences occurred in VTE rates (6.16 vs. 6.23, p=0.8) or bleeding (0.23 vs. 0.33, p=0.084) in the propensity-matched cohort. VTE and bleeding events were both identified using ICD-9 coding (Pharmacotherapy 2019;39:232-41).
   3. Bioavailability of subcutaneously administered drugs is reduced in critically ill patients with the concomitant use of vasoactive drugs or the presence of edema, thereby potentially reducing the effect.
   4. A high proportion of critically ill patients have acute kidney injury, which may limit the use of low-molecular-weight heparin. Critically ill patients (n=138) administered prophylactic subcutaneous dalteparin with an estimated creatinine clearance (CrCl) of less than 30 mL/minute/1.73 $m^2$ were evaluated in a prospective study. No evidence was found of accumulation or an increased risk of bleeding (Arch Intern Med 2008;168:1805-12).
   5. Bleeding
      a. Bleeding rates in critically ill patients vary, depending on the type of pharmacologic prophylaxis.
      b. Patients at high risk of bleeding are often excluded from studies.
      c. Patients at high risk of bleeding with a moderate to high risk of VTE may be considered for mechanical VTE prophylaxis; however, pharmacologic prophylaxis should be reassessed when the bleeding risk is no longer present.
   6. Limited evidence exists to guide dosing in the critically ill population with obesity. An inverse relationship between body weight and anti-factor Xa (anti-Xa) concentration may exist in patients with obesity; however, the risk of VTE and the optimal anti-Xa concentrations to achieve are unclear. Some clinicians may choose heparin 7500 units subcutaneously every 8 hours over heparin 5000 units subcutaneously every 8 hours in patients with BMI greater than 40 kg/$m^2$; however, several studies, which were limited by sample size, have failed to demonstrate a clinical benefit (Pharmacotherapy 2016;36:740-8; Hosp Pharm 2016;51:376-81). There is also controversy regarding the optimal approach to VTE prophylaxis with enoxaparin in patients with a BMI greater than 40 kg/$m^2$; approaches to dosing and anti-Xa level vary significantly because of the varying populations studied. Common

strategies include either a fixed dose of 40 mg subcutaneously twice daily or 0.5 mg/kg twice daily. For patients with a BMI greater than 50 mg/m$^2$, in addition to weight-based dosing, some clinicians use 60 mg subcutaneously twice daily (Ann Pharmacother 2011;45:1356-62; J Pharm Technol 2015;31:282-8).

F. Oral Anticoagulants for VTE Prophylaxis
   1. The only study evaluating direct oral anticoagulants (DOACs) for VTE prophylaxis in critically ill patients was a substudy of extended duration (35-42 days) of betrixaban versus shorter duration (10 ± 4 days) low-molecular-weight heparin in critically ill medical patients. Of the 7513 patients in the full cohort, 703 critically ill patients were included in the substudy. Extended-duration betrixaban reduced the rate of VTE at 35-42 days from 7.95% to 4.27% (p=0.042). Major bleeding in the two groups was 1.14% and 3.13% (p=0.07).
      a. These findings should currently be treated as exploratory, given that the full study had predefined statistical stipulations that were unmet (Intensive Care Med 2019;45:477-87).
      b. Betrixaban was removed from the market in April 2020 because of a lack of commercial success.
   2. Rivaroxaban is noninferior to standard treatments in other settings such as orthopedic surgery.
   3. Rivaroxaban 10 mg orally once daily was compared with enoxaparin 40 mg subcutaneously daily for 10 days, followed by placebo in acutely ill hospitalized patients. The rates of asymptomatic or symptomatic VTE, pulmonary embolism, or death were comparable; however, the bleeding rates were increased in the rivaroxaban group (N Engl J Med 2013;368:513-23). A subsequent study evaluating rivaroxaban 10 mg/day orally with placebo in patients at high risk of VTE demonstrated a decrease in rates of symptomatic VTE (0.18 vs 0.42%; HR 0.44, 95% CI, 0.22-0.89); however, low overall VTE and bleeding event rates limit widespread use. Further studies must identify the populations most likely to benefit from extended prophylaxis (N Engl J Med. 2018;379:1118-27).
   4. Low-molecular-weight heparin is preferred to vitamin K antagonists such as warfarin and DOACs for prophylaxis (Blood Adv 2018;2:3360-92).

G. Heparin-Induced Thrombocytopenia (Chest 2012;141(2 suppl):495S-530S)
   1. HIT is a severe, immune-mediated reaction that potentially leads to life-threatening complications such as myocardial infarction, skin necrosis, stroke, and VTE (around 50%–75% of patients with HIT develop symptomatic thrombosis).
   2. A rare manifestation is delayed-onset HIT, affecting patients exposed to heparin in the recent past (prior 2 weeks) who present with a new thrombosis and low platelet count.
   3. Frequency of HIT
      a. Higher in patients receiving unfractionated heparin than in patients receiving low-molecular-weight heparin, occurring in 1%–5% of patients versus less than 1%, respectively
      b. Occurs in less than 1% of ICU patients
      c. Higher risk in cardiac or orthopedic surgical patients receiving unfractionated heparin (15%) than in medical patients (0.1%–1%)
   4. Alternative causes of thrombocytopenia in critically ill patients include extracorporeal devices, intra-aortic balloon pumps, sepsis, disseminated intravascular coagulation, bleeding, and medications. Platelet counts may decrease after surgery on cardiopulmonary bypass and subsequently recover; however, a secondary decrease in platelet count may signal potential HIT.
   5. Clinical diagnosis of HIT
      a. Suspected when a patient has a decrease in absolute platelet count to less than 150,000/mm$^3$ or a relative decrease of at least 50% from baseline, skin lesions at injection sites, or systemic reactions after intravenous boluses
      b. Typical onset is 5–10 days after heparin exposure, though onset can be delayed and can occur up to 3 weeks after therapy cessation.

c. Recent heparin exposure, within the last 90 days, may result in rapid-onset HIT, occurring within hours after rechallenge.
d. Patients with recent unfractionated heparin/low-molecular-weight heparin exposure and a new thrombosis should have their platelet count checked before anticoagulant therapy is initiated.

6. Probability of HIT
   a. 4T score
      i. Segregates patients into low, intermediate, and high clinical probability on the basis of four criteria
      ii. Four clinical features incorporated: (1) thrombocytopenia, (2) timing of thrombocytopenia, (3) presence of thrombosis or other clinical sequelae, and (4) other causes of thrombocytopenia
      iii. A low probability 4T score has a high negative predictive value (0.998), whereas an intermediate or high probability score has a low positive predictive value (0.64 and 0.14, respectively) (Blood 2012;120:4160-7).
   b. HEP (HIT Expert Probability) score
      i. Incorporates more clinical features than the 4T score
      ii. More complex (time-consuming) and evaluation studies limited (not in ICU patients)
7. Laboratory testing
   a. Antigen assays
      i. Depends on the detection of antibody binding by enzyme-linked immunosorbent assays or particle-based immunoassays
      ii. A recently approved automated latex immunoturbidimetric assay has been made available. This assay does not require sample batching as is common with the enzyme-Linked immunosorbent assays; thus, results can be obtained within 20 minutes.
      iii. Antibody present if sample from patient binds to the heparin-PF4–coated wells, leading to a color-producing reaction. A higher antibody concentration leads to greater color production and a higher optical density reading. Optical density readings of 0.4 or greater are considered positive and indicative of HIT antibodies.
      iv. High sensitivity (greater than 90%) and low to moderate specificity
         (a) Clinically insignificant HIT antibodies are often detected among patients who received heparin 5–100 days earlier.
         (b) Detects a range of immunoglobulin (Ig) A and IgM antibodies that are not pathogenic
   b. Functional assays
      i. Examples: Heparin-induced platelet aggregation (HIPA) and C14 serotonin release assay
      ii. Detect platelet activation in the presence of heparin. Patient serum is mixed with washed platelets from healthy volunteers and low and high concentrations of heparin. In the presence of HIT antibodies, platelets are activated in low concentrations of heparin and detected using radioactive serotonin (serotonin release assay) or visually (HIPA).
      iii. High sensitivity and specificity
      iv. Technically challenging and not readily available
8. Treatment of HIT (Table 10)
   a. An intermediate or high 4T score should prompt immediate discontinuation of all sources of heparin and initiation of an alternative nonheparin anticoagulant until confirmatory testing is completed.
   b. Parenteral direct thrombin inhibitors are the agents of choice for anticoagulation in acute HIT because they have no cross-reactivity with heparin. Some studies support the use of the factor Xa inhibitor fondaparinux for the treatment of HIT, though there are reports of fondaparinux-induced HIT.

c. Parenteral direct thrombin inhibitors are associated with a higher rate of major bleeding complications than is unfractionated heparin.
d. Initiate warfarin once the platelet count has recovered and is within normal limits (at least 150,000/mm$^3$) and after at least 5 days of therapy with an alternative parenteral anticoagulant. Alternatively, conservative warfarin dosing may begin once the platelet count is recovering. If a patient is receiving warfarin at the time of HIT diagnosis, reversing with vitamin K is recommended.
e. DOACs are another option that significantly simplify HIT management. Currently, data analyses supporting DOACs for HIT are limited to case series and small observational cohort studies (Thromb Res 2015;135:607-9; Blood 2017;130:1104-13; Am J Cardiovasc Drugs 2022;22:417-24). DOACs are a recommended therapeutic option in the American Society of Hematology 2018 guidelines for VTE management in HIT.
    i. DOAC selection should be based on patient- and drug-specific factors
    ii. Varying dosing strategies have been described based on the presence of thrombosis and/or platelet recovery
f. Argatroban dosing in the critically ill population (Crit Care 2010;14:R90; Ann Pharmacother 2007;41:749-54)
    i. Mean dose in critically ill patients was 0.24 ± 0.16 mcg/kg/minute and was 0.22 ± 0.15 mcg/kg/minute in critically ill patients with multiple organ dysfunction.
    ii. In patients with severe liver impairment, consider 0.5 mcg/kg/minute.
    iii. The target aPTT is 1.5–3 times baseline.
g. Bivalirudin dosing in the critically ill population (Pharmacotherapy 2006;26:452-60)
    i. Dose reduced to 0.05–0.1 mg/kg/hour, depending on renal function and bleeding risks
    ii. The target aPTT is 1.5–2.5 times baseline.

**Table 10.** Parenteral Agents for the Treatment of HIT

| | Argatroban | Bivalirudin | Fondaparinux |
|---|---|---|---|
| FDA approved for the treatment of HIT | Yes | Yes (percutaneous coronary intervention with HIT) | No |
| Mechanism of action | Direct thrombin inhibitor | Direct thrombin inhibitor | Factor Xa inhibitor |
| Elimination half-life | 40–50 min | 25 min | 17–20 hr |
| Elimination | Hepatobiliary | 80% enzymatic<br>20% renal | Renal |
| Dosing | 2 mcg/kg/min<br>(see above for dosing in the critically ill population) | Unlabeled dose for HIT:<br>0.15–0.2 mg/kg/hr<br>(see above for dosing in the critically ill population) | Unlabeled dose for HIT: 5–10 mg SC daily (depending on weight); 2.5 mg/day for VTE prophylaxis |
| Monitoring | aPTT | aPTT | Anti-Xa concentration |
| INR elevation | Excessive | Moderate | None |

aPTT = activated partial thromboplastin time; HIT = heparin-induced thrombocytopenia.

**Patient Cases**

5. A 93-year-old man (weight 45 kg) confined to his bed is admitted from a nursing home with a chronic obstructive pulmonary disease exacerbation requiring mechanical ventilation. He has a history of diabetes and heart failure. His laboratory values are all within normal limits except for BUN 35 mg/dL and SCr 2.8 mg/dL (baseline 0.5). Which would be the most appropriate recommendation for VTE prophylaxis in this patient?
   A. Intermittent pneumatic compression devices.
   B. Enoxaparin 30 mg subcutaneously once daily.
   C. Heparin 5000 units subcutaneously twice daily.
   D. Fondaparinux 2.5 mg subcutaneously daily.

*Questions 6 and 7 pertain to the following case*.
A 55-year-old man (weight 60 kg) with a medical history of diabetes, hyperlipidemia, and a DVT 4 months ago secondary to lower-extremity trauma is admitted today to the ICU for acute respiratory failure from influenza virus. His current laboratory values are as follows: WBC 13.1 x $10^3$ cells/$mm^3$, Plt 250,000/$mm^3$, BUN 13 mg/dL, SCr 0.9 mg/dL, INR 1.2, AST 22 IU/mL, and ALT 11 IU/mL. His current medication regimen includes fentanyl and midazolam boluses for pain and agitation, piperacillin/tazobactam, vancomycin, regular insulin infusion, SUP, and a heparin drip. Five days later, the patient remains intubated on the same medications. At this time, his Plt has decreased to 112,000/$mm^3$, and his BUN and SCr have increased to 45 mg/dL and 2.7 mg/dL, respectively. The team sends a heparin-PF4 immunoassay; however, the results will not return for 48 hours.

6. Which would be the best course of action?
   A. Discontinue the heparin drip, and initiate an argatroban continuous infusion at 0.25 mcg/kg/minute.
   B. Do nothing because the patient has several other reasons to be thrombocytopenic.
   C. Discontinue the heparin drip, and initiate fondaparinux at 10 mg subcutaneously daily.
   D. Do nothing until the heparin-PF4 immunoassay results return.

7. Three days later, both the heparin-PF4 immunoassay and the serotonin release assay return positive, and the patient has a new DVT. The team wants to initiate warfarin. The patient's current Plt is 130,000/$mm^3$. Which would be the most appropriate response?
   A. Discontinue argatroban and initiate warfarin at 5 mg orally daily.
   B. Discontinue argatroban and initiate warfarin at 10 mg orally daily.
   C. Warfarin should never be used in patients with HIT.
   D. Warfarin should not be initiated right now.

## IV. END-OF-LIFE CARE

A. Clinicians commonly provide end-of-life and palliative care in ICUs.

B. The World Health Organization describes palliative care as "an approach that improves the quality of life of patients and their families facing the problems associated with life-threatening illness, through the prevention and relief of suffering by means of early identification and impeccable assessment and treatment of pain and other problems, physical, psychosocial, and spiritual" (Global Atlas of Palliative Care at the End of Life).

C. Goals of Palliative Care
   1. Provides relief from pain and other distressing symptoms.
   2. Affirms life and regards dying as a normal process.
   3. Intends neither to hasten nor postpone death.
   4. Integrates the psychological and spiritual aspects of patient care.
   5. Offers a support system to help patients live as actively as possible until death.
   6. Offers a support system to help the family cope during the patient's illness and in their own bereavement; uses a team approach to address the needs of patients and their families, including bereavement counseling, if indicated.
   7. Enhances quality of life, and may also positively influence the course of illness.
   8. Is applicable early in the course of illness, in conjunction with other therapies that are intended to prolong life, such as chemotherapy or radiation therapy, and includes those investigations needed to better understand and manage distressing clinical complications.

D. Categories of Support
   1. Pain management is of paramount importance for comfort and reduction of distress. Providers and families can collaborate to identify the sources of pain and relieve them with drugs and other forms of therapy.
   2. Symptom management involves treating symptoms other than pain such as nausea, thirst, bowel and bladder problems, depression, anxiety, dyspnea, and secretions.
   3. Emotional and spiritual support is important for both the patient and the family in dealing with the emotional demands of critical illness.

E. General Considerations
   1. Minimize or discontinue the use of uncomfortable or unnecessary procedures, tests, or treatments.
   2. Minimize or discontinue the use of routine vital sign checks, patient weights, cardiac or other electronic monitoring, fingersticks, and intermittent pneumatic compression devices.
   3. Consider discontinuing routine blood tests, radiologic imaging, and other diagnostic procedures.
   4. Consider discontinuing all medications not necessary for patient comfort.
   5. Neuromuscular blocking agents should be discontinued, and their effects allowed to reverse, to best assess the patient's comfort level before withdrawal of life support. If this is not possible due to the delay that paralysis cessation would contribute to withdrawal of life support, best efforts should be made to detect discomfort.

F. Symptom Management
   1. Pain
      a. No evidence supports that unconscious patients do not experience pain.
      b. Opioids are the treatment mainstay for patients with pain at the end of life.
      c. Administer an opioid as an intravenous bolus dose, and begin an intravenous continuous infusion, adjusting rates to maintain comfort; avoid using subcutaneous or enteral, unless intravenous access is unavailable, because the onset is delayed and absorption may be unpredictable (Crit Care Med 2019;47:1619-26).
      d. Bolus and titrate infusion to control labored respirations; specific dosages of medications are less important than the goal of symptom relief. Optimal dose is determined by assessing the patient and rapidly increasing the dose as needed until symptoms are no longer present. Dose is determined by symptom relief and adverse effects (excessive sedation, respiratory depression).

e. Suggested goals include keeping the respiratory rate at or below 30 breaths/minute and keeping the patient pain free. Pain assessment should include using the visual analog scale, the behavioral pain scale, or the Critical-Care Pain Observation Tool (see the Management of Pain, Agitation, Delirium and Neuromuscular Blockade in Adult Intensive Care Unit Patients chapter for further details on these scales).
f. Never use neuromuscular blocking agents to treat pain.
g. Morphine is most commonly used; hydromorphone and fentanyl are alternatives.
h. In addition, opioids reduce dyspnea.
i. Tolerance may develop over time.
j. Pain can be improved with correct dosing and titration without causing respiratory depression or hastening death (Chest 2004;126:286-93; Crit Care Med 2004;32:1141-8; JAMA 1992;267:949-53).

2. Dyspnea
   a. Common symptom in patients at the end of life
   b. Individualize therapy based on underlying cause, patient's level of consciousness, and level of sedation.
   c. Oxygen may be used for patients with hypoxia. Corticosteroids, bronchodilators and diuretics may also be useful (Crit Care Med 2008;36:953-63).
   d. Opioids are the first-line therapy. Opioids reduce oxygen consumption, ventilation, and perception of dyspnea (J Palliat Med 2012;15:106-14).
   e. No benefit with benzodiazepines unless anxiety is present (Cochrane Database Syst Rev 2010;1:CD007354)
3. Anxiety/agitation/delirium
   a. Symptoms at the end of life can relate to acute or chronic anxiety, delirium, or terminal delirium.
   b. Nonpharmacologic treatments for agitation and anxiety can include frequent reorientation to the environment and reduction in noise and other bothersome or stimulating environmental factors.
   c. Haloperidol is the agent of choice for delirium because of its proven efficacy, the availability of an intravenous injection, and the agent's less sedative effects compared with other agents (Crit Care Med 2008;36:953-63). Intravenous haloperidol may be used without electrocardiographic (ECG) monitoring because the benefits outweigh the risks of prolonged corrected QT interval, given the goals of care.
   d. Benzodiazepines (midazolam and lorazepam):
      i. Benzodiazepines are the agents of choice for anxiety. Dose is determined by assessing the patient and increasing the dose as needed (lower initial doses and titration with frequent assessment).
      ii. Determining what would be perceived as an acceptable level of sedation with the patient and/or family or surrogate decision-maker is important before initiating sedatives.
      iii. Tolerance may develop over time.
4. Fever
   a. Acetaminophen is an effective therapy for improving comfort and decreasing the incidence of fever. If the patient cannot swallow, this agent may be administered per rectum. If neither enteral nor rectal access are available, acetaminophen may be administered intravenously.
   b. A nonsteroidal anti-inflammatory drug may be used when acetaminophen is ineffective.
   c. Dexamethasone, which is also known to have antipyretic properties, can be considered.
5. Nausea and vomiting
   a. Underlying causes such as medications, uremia, ascites, gastroparesis, and intestinal or gastric obstruction should be treated or eliminated, if possible.
   b. Agents to consider include metoclopramide, haloperidol, risperidone, ondansetron, and dexamethasone.
   c. Lorazepam can be considered as an adjunct, especially with anticipatory vomiting.
   d. Use of more than one agent from different classes may be necessary for symptom relief.

6. Cough
    a. Excessive coughing can lead to exacerbation of dyspnea and spells of nausea and vomiting, in addition to disturbing sleep and exacerbating pain.
    b. Non-opioid antitussives such as benzonatate and dextromethorphan may be considered.
    c. All opioids have intrinsic antitussive action by inhibiting the brain stem cough center; however, if the patient is receiving an opioid for other reasons, adding another opioid has not shown additional benefit.
    d. For refractory cough, consider nebulized lidocaine.
7. Secretions
    a. Near the end of life, the ability to clear oral and tracheobronchial secretions diminishes.
    b. Secretions are usually too low in the tracheobronchial tree for gentle oral suctioning to help, and suctioning can be disturbing.
    c. The treatment mainstay includes anticholinergic and antimuscarinic medications.
        i. Glycopyrrolate (0.1 mg intravenously every 4 hours) or atropine (1% ophthalmic solution 2 drops sublingually every 4 hours as needed) should be used for acute symptoms.
        ii. Scopolamine and atropine cross the blood-brain barrier and can be more sedating than glycopyrrolate.
        iii. The scopolamine patch is more gradual in onset (12 hours).
        iv. More than 1 scopolamine patch may be used for unrelieved symptoms.

**Patient Case**

8. An 88-year-old woman is admitted to the ICU for decompensated heart failure, acute kidney injury, and uncontrollable pain from the rib fractures she had 1 month ago from a fall. This is her fourth admission to the ICU in the past 5 months. Speaking with her, you find that she wishes not to be resuscitated or intubated but only to be comfortable. Her blood pressure is currently 119/70 mm Hg, heart rate 120 beats/ minute, and respiratory rate 55 breaths/minute. Her pain is 9/10 using the BPS. In a meeting with the patient's family, all members agree that they do not want to see her suffer any longer. It is decided to initiate a morphine drip at 2 mg/hour. Titration parameters include giving a bolus dose equivalent to the current rate and increasing the infusion by 25% to maintain a score of 3 (no pain) using the BPS. Her laboratory values are all within normal limits, including BUN 10 mg/dL and SCr 0.6 mg/dL. The nurse taking care of the patient believes that the titration parameters are too aggressive. Which would be the most appropriate change in titration parameters?

    A. Change the parameters to increase the morphine drip when the patient has signs of discomfort, such as an increase in blood pressure or heart rate.
    B. Discontinue titration parameters, keeping the morphine infusion at the current rate.
    C. Discontinue titration parameters, keeping the morphine infusion at the current rate and adding a midazolam infusion at 2 mg/hour.
    D. Do not change the titration parameters at this time; however, assess the patient's response after the first dose increase.

## V. DISASTER MANAGEMENT

A. The Federal Emergency Management Association (FEMA) defines a *disaster* as any occurrence that results in property damage, deaths or injuries to a community; this occurrence encompasses events ranging from natural or weather events, infectious outbreaks, chemical or radiation releases, terrorist attacks or threats, mass violence, transportation accidents, to technological disasters (https://training.fema.gov/programs/emischool/el361toolkit/glossary.htm)

B. Disaster planning often centers on emergency management; however, a critical care disaster management plan is critical to supporting the influx of critically ill patients who may present for care because of such events (Crit Care Clin.2019 Oct;35(4):551-562.)
   1. Emergency management consists of 4 phases
      a. Preparedness
      b. Response
      c. Mitigation
      d. Recovery
   2. Disaster risk is dependent on the geographic area and institution; a hazard vulnerability analysis (HVA) should be completed annually to identify disaster risks and prioritize mitigation strategies (CDC Hospital All-Hazards Self-Assessment, www.cdc.gov/orr/readiness/resources/healthcare/documents/hah_508_compliant_final.pdf)
   3. Many resources are available to assist in the event of disaster
      a. National Incident Management System (NIMS)—provides guidance to private and public sectors on resource management, command and coordination, and communications and information management for disasters
      b. U.S. States Department of Homeland Security (DHS)
      c. U.S. Department of Health and Human Services (HHS)
         i. Operates the Strategic National Stockpile (SNS), which stores medications and supplies throughout the country to enable rapid dissemination on federal instruction; stocked medications include antibiotics, antivirals, chemical antidotes, antitoxins, vaccines
         ii. Appoints the Assistance Secretary for Preparedness and Response (ASPR) who creates the Technical Resources, Assistance Center and Information Exchange (TRACIE), which contains recommendations for many disasters situations
      d. Local and State Public Health Departments—provide disaster response training for healthcare providers
      e. FEMA—coordinates disaster response in events where local and state response by be overwhelmed
   4. An interprofessional approach to disaster response is key; pharmacists are essential in many elements (Am J Health Syst Pharm 2004;61:1167-75)
      a. Response integration
         i. Ensure appropriate use of medications
         ii. Recognition of bioterrorism events and associated treatments
         iii. First aid and advanced cardiovascular life support skills
         iv. Patient triage and counseling
         v. Interdisciplinary education
      b. Patient management
         i. Drug therapy selection and management
         ii. Patient education regarding therapies, discouraging personal stockpiles and self-treatment
         iii. Ensure continuity of care for chronic disease states via collaborative practice agreements
         iv. Point of care testing and treatment

v. Vaccine administration
vi. Telehealth and telepharmacy services

c. Medication supply
i. Identify necessary antidotes, critical medications and alternatives, as well as necessary par levels
ii. Ensure proper preparation and handling of medications
iii. Maintain effective distribution systems
iv. Compile accurate therapeutic and patient records

d. Policy coordination
i. Develop diagnosis and treatment guidelines and/or policies
ii. Coordinate efforts with local and state health departments
iii. Incorporate medication expertise in policy decisions at local and state level

e. While pharmacists are well-suited to perform the aforementioned tasks, in a disaster response it is also necessary to consider which tasks would most benefit from pharmacist input and which tasks may be delegated to other healthcare professionals

## VI. SOCIAL DETERMINANTS OF HEALTH

A. Social determinants of health (SDoH) are environmental conditions that impact morbidity, mortality, and quality-of-life. SDoH encompass five domains of modifiable and non-modifiable factors (Healthy People 2030, HHS, Office of Disease Prevention and Health Promotion, https://health.gov/healthypeople):
1. Economic stability—employment, food insecurity, housing instability, poverty
2. Education access and quality—early childhood development and education, higher education enrollment, high school graduation, language and literary
3. Health care access and quality—access to health services and primary care, health literacy
4. Neighborhood and built environment—access to foods that support a healthy diet, crime and violence, environmental conditions, housing quality
5. Social and community context—civic participation, discrimination, incarceration, social cohesion, culture, family

B. Recognizing and addressing SDoH in critically ill patients has the potential to positively impact health care outcomes and costs by identifying opportunities for societal improvement (HHS, Office of the Assistant Secretary for Planning and Evaluation. Building the Evidence Base for Social Determinants of Health Interventions, https://aspe.hhs.gov/reports/building-evidence-base-social-determinants-health-interventions)
1. Some professional societies, such as the American Thoracic Society and American Heart Association, recommend training of clinicians to facilitate patients screening for SDoH; however, formal curricula and screening tools are lacking (Ann Am Thorac Soc 2017;14:814-26)
2. Engaging patients from various backgrounds in research is also critical for identifying barriers to care and potential interventions to address these gaps, as well as to identify other population differences
3. ICU recovery clinics are in an excellent position to pave the way towards addressing SDoH by engaging with patients and caregivers about the challenges they face in post-ICU care

C. Studies evaluating SDoH in the critically ill population are limited and often rely on claims data for their conclusions; it is often difficult to identify the specific SDoH associated with a particular outcomes because different SDoH are tied closely together

1. A retrospective evaluation of older adults admitted to the ICU demonstrated a greater risk for functional and cognitive decline in patients with dual Medicare–Medicaid coverage compared with those with Medicare with supplemental insurance, suggesting opportunities for improved allocation of rehabilitation resources and post-ICU care (Ann Intern Med 2022;175:644-55). Other studies reinforce this finding, indicating uninsured patients are less likely to receive common critical care procedures (Am J Respir Crit Care Med 2011;184:809-15).
2. Neighborhood socioeconomic disadvantage and lower socioeconomic status have similarly been associated with higher disability burden after ICU discharge (Chest 2016;150:829-36; Crit Care Med 2022;50:733-41; Intensive Care Med 2018;44:2025-37)
3. Lower individual socioeconomic status has also been associated with increased health care use and mortality (Crit Care Med 2019; 47:e512-21; J Intensive Care Med 2021;36:828-37)
4. Members of minority groups are demonstrated to experience higher mortality and disparate end of life care (e.g., less likely to use hospice, fewer advanced directives, less comfort-directed care) (Lancet Respir Med 2017;5:E11-2; Ann Am Thorac Soc 2016;13:2184-9)

## REFERENCES


**Key Aspects in the General Care of All Critically Ill Patients**

1. Haynes AB, Weiser TG, Berry WR, et al. A surgical safety checklist to reduce morbidity and mortality in a global population. N Engl J Med 2009;360:491-9.
2. Mabasa VH, Malyuk DL, Weatherby EM, et al. A standardized, structured approach to identifying drug-related problems in the intensive care unit: FASTHUG-MAIDENS. Can J Hosp Pharm 2011;64:366-9.
3. Pronovost P, Needgam D, Berenholtz S, et al. An intervention to decrease catheter-related bloodstream infections in the ICU. N Engl J Med 2006;355:2725-32.
4. Rhodes A, Evans LE, Alhazzani W, et al. Surviving Sepsis Campaign: international guidelines for management of sepsis and septic shock: 2016. Crit Care Med 2017;44:486-552.
5. Vincent JL. Give your patient a fast hug (at least) once a day. Crit Care Med 2005;33:1225-9.
6. Vincent WR III, Hatton KW. Critically ill patients need "FAST HUGS BID" (an updated mnemonic). Crit Care Med 2009;37:2326-7; author reply 2327.

**Stress Ulcer Prophylaxis**

1. Alhazzani W, Alenezi F, Jaeschke RZ, et al. Proton pump inhibitors versus histamine 2 receptor antagonists for stress ulcer prophylaxis in critically ill patients: a systematic review and meta-analysis. Crit Care Med 2013;41:693-705.
2. Alhazzani W, Alshamsi F, Belley-Cote E, et al. Efficacy and safety of stress ulcer prophylaxis in critically ill patients: a network meta-analysis of randomized trials. Intensive Care Med 2018;44:1-11.
3. ASHP Therapeutic Guidelines on Stress Ulcer Prophylaxis. Am J Health Syst Pharm 1999;56:347-79.
4. Barkun AN, Adam V, Martel M, et al. Cost-effectiveness analysis: stress ulcer bleeding prophylaxis with proton pump inhibitors, H2 receptor antagonists. Value Health 2013;16:14-22.
5. Barkun AN, Bardou M, Pham CQ, et al. Proton pump inhibitors vs. histamine 2 receptor antagonists for stress-related mucosal bleeding prophylaxis in critically ill patients: a meta-analysis. Am J Gastroenterol 2012;107:507-20.
6. Bateman BT, Bykov K, Choudhry NK, et al. Type of stress ulcer prophylaxis and risk of nosocomial pneumonia in cardiac surgical patients: cohort study. BMJ 2013;347:f5416.
7. Buendgens L, Bruensing J, Matthes M, et al. Administration of proton pump inhibitors in critically ill medical patients is associated with increased risk of developing *Clostridium difficile*-associated diarrhea. J Crit Care 2014;29:696.
8. Cook D, Guyatt G, Marshall J, et al. A comparison of sucralfate and ranitidine for the prevention of upper gastrointestinal bleeding in patients requiring mechanical ventilation. Canadian Critical Care Trials Group. N Engl J Med 1998;338:791-7.
9. Cook D, Heyland D, Griffith L, et al. Risk factors for clinically important upper gastrointestinal bleeding in patients requiring mechanical ventilation. Canadian Critical Care Trials Group. Crit Care Med 1999;27:2812-7.
10. Cook DJ, Fuller HD, Guyatt GH, et al. Risk factors for gastrointestinal bleeding in critically ill patients. Canadian Critical Care Trials Group. N Engl J Med 1994;330:377-81.
11. El-Kersh K, Jalil B, McClave SA, et al. Enteral nutrition as stress ulcer prophylaxis in critically ill patients: a randomized controlled exploratory study. J Crit Care 2018;43:108-13.
12. Guillamondegui OD, Gunter OL, Bonadies JA, et al. Practice Management Guidelines for Stress Ulcer Prophylaxis 2008. Available at www.east.org.
13. Huang HB, Jiang W, Wang CY, et al. Stress ulcer prophylaxis in intensive care unit patients receiving enteral nutrition: a systematic review and meta-analysis. Crit Care 2018;22:20.
14. Krag M, Marker S, Perner A. Pantoprazole in patients at risk for gastrointestinal bleeding in the ICU. N Engl J Med 2018;379:2199-208.
15. Krag M, Perner A, Wetterslev J, et al. Prevalence and outcome of gastrointestinal bleeding and use of acid suppressants in acutely ill adult intensive care patients. Intensive Care Med 2015;41:833-45.
16. Lin P, Chang C, Hsu P, et al. The efficacy and safety of proton pump inhibitors vs histamine-2 receptor

antagonists for stress ulcer bleeding prophylaxis among critical care patients: a meta-analysis. Crit Care Med 2010;38:1197-205.
17. MacLaren R, Campbell J. Cost-effectiveness of histamine receptor-2 antagonist versus proton pump inhibitor for stress ulcer prophylaxis in critically ill patients. Crit Care Med 2014;42:809-15.
18. MacLaren R, Reynolds PM, Allen RR. Histamine-2 receptor antagonists vs proton pump inhibitors on gastrointestinal tract hemorrhage and infectious complications in the intensive care unit. JAMA Intern Med 2014;174:564-74.
19. Madsen KR, Lorentzen K, Clausen N, et al. Guideline for stress ulcer prophylaxis in the intensive care unit. Dan Med J 2014;61:C4811.
20. Marik PE, Vasu T, Hirani A, et al. Stress ulcer prophylaxis in the new millennium: a systematic review and meta-analysis. Crit Care Med 2010;38:2222-8.
21. McRorie JW, Kirby JA, Miner PB. Histamine2-receptor antagonists: rapid development of tachyphylaxis with repeat dosing. World J Gastrointest Pharmacol Ther 2014;5:57-62.
22. PEPTIC Investigators for the Australian and New Zealand Intensive Care Society Clinical Trials Group, Alberta Health Services Critical Care Strategic Clinical Network, and the Irish Critical Care Trials Group; Young PJ, Bagshaw SM, Forbes AB, et al. Effect of stress ulcer prophylaxis with proton pump inhibitors vs histamine-2 receptor blockers on in-hospital mortality among ICU patients receiving invasive mechanical ventilation: the PEPTIC Randomized Clinical Trial. JAMA 2020;323:616-26.
23. Rhodes A, Evans LE, Alhazzani W, et al. Surviving Sepsis Campaign: international guidelines for management of sepsis and septic shock: 2016. Crit Care Med 2017;44:486-552.
24. Simons RK, Hoyt DB, Winchell RJ, et al. A risk analysis of stress ulceration after trauma. J Trauma 1995;39:289-93.
25. Ye Z, Reintam Blaser A, Lytvyn L, et al. Gastrointestinal bleeding prophylaxis for critically ill patients: a clinical practice guideline. BMJ 2020;368:l6722.

**Prophylaxis Against Deep Venous Thrombosis or Pulmonary Embolism**

1. Arabi YM, Al-Hameed F, Burns KEA, et al. Adjunctive intermittent pneumatic compression for venous thromboprophylaxis. N Engl J Med 2019;380:1305-15.
2. Attia J, Ray JG, Cook DJ, et al. Deep vein thrombosis and its prevention in critically ill adults. Arch Intern Med 2001;161:1268-79.
3. Beiderlinden M, Treschan TA, Görlinger K, et al. Argatroban anticoagulation in critically ill patients. Ann Pharmacother 2007;41:749-54.
4. Cade JF. High risk of the critically ill for venous thromboembolism. Crit Care Med 1982;10:448-50.
5. Chi G, Gibson CM, Kalayci A, et al. Extended-duration betrixaban versus shorter-duration enoxaparin for venous thromboembolism prophylaxis in critically ill medical patients: an APEX trial substudy. Intensive Care Med 2019;45:477-87.
6. Cohen AT, Spiro TE, Büller HR, et al. Rivaroxaban for thromboprophylaxis in acutely ill medical patients. N Engl J Med 2013;368:513-23.
7. Cook D, Crowther M, Meade M, et al. Deep venous thrombosis in medical-surgical critically ill patients: prevalence, incidence, and risk factors. Crit Care Med 2005;33:1565-71.
8. Cook D, Meade M, Guyatt G, et al. Dalteparin versus unfractionated heparin in critically ill patients. N Engl J Med 2011;364:1305-14.
9. Cuker A, Arepally GM, Chong BH. American Society of Hematology 2018 guidelines for management of venous thromboembolism: heparin-induced thrombocytopenia. Blood Adv 2018;2:3360-92.
10. Cuker A, Gimotty PA, Crowther MA, et al. Predictive value of the 4Ts scoring system for diagnosis and management of heparin-induced thrombocytopenia: a systemic review and meta-analysis. Blood 2012;120:4160-7.
11. De A, Roy P, Garg VK, et al. Low-molecular-weight heparin and unfractionated heparin in prophylaxis against deep vein thrombosis in critically ill patients undergoing major surgery. Blood Coagul Fibrinolysis 2010;21:57-61.

12. Douketis J, Cook D, Meade M, et al. Prophylaxis against deep vein thrombosis in critically ill patients with severe renal insufficiency with the low-molecular-weight heparin dalteparin: an assessment of safety and pharmacodynamics: the DIRECT study. Arch Intern Med 2008;168:1805-12.
13. Fraisse F, Holzapfel L, Couland JM, et al. Nadroparin in the prevention of deep vein thrombosis in acute decompensated COPD. The Association of Non-University Affiliated Intensive Care Specialist Physicians of France. Am J Respir Crit Care Med 2000;161:1109-14.
14. Geerts W, Cook D, Selby R, et al. Venous thromboembolism and its prevention in critical care. J Crit Care 2002;17:95-104.
15. Gould MK, Garcia DA, Wren SM, et al. Prevention of VTE in nonorthopedic surgical patients: Antithrombotic Therapy and Prevention of Thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. Chest 2012;141(2 suppl):227S-77S.
16. Kahn SR, Lim W, Dunn AS, et al. Prevention of VTE in nonsurgical patients: Antithrombotic Therapy and Prevention of Thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. Chest 2012;141(2 suppl):195S-226S.
17. Kiser TH, Fish DN. Evaluation of bivalirudin treatment for heparin-induced thrombocytopenia in critically ill patients with hepatic and/or renal dysfunction. Pharmacotherapy 2006;26:452-60.
18. Linkins LA, Dans AL, Moores LK, et al. Treatment and prevention of heparin-induced thrombocytopenia: Antithrombotic Therapy and Prevention of Thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. Chest 2012;141(2 suppl):495S-530S.
19. Phung OJ, Kahn SR, Cook DJ, et al. Dosing frequency of unfractionated heparin thromboprophylaxis: a meta-analysis. Chest 2011;140:374-81.
20. Reynolds PM, Van Matre ET, Wright GC, et al. Evaluation of prophylactic heparin dosage strategies and risk factors for venous thromboembolism in the critically ill patient. Pharmacotherapy 2019;39:232-41.
21. Sachdeva A, Dalton M, Amaragiri SV, et al. Graduated compression stockings for prevention of deep vein thrombosis. Cochrane Database Syst Rev 2014;12:1-72.
22. Saugel B, Phillip V, Moessmer G, et al. Argatroban therapy for heparin-induced thrombocytopenia in ICU patients with multiple organ dysfunction syndrome: a retrospective study. Crit Care 2010;14:R90.
23. Shorr AF, Williams MD. Venous thromboembolism in critically ill patients: observations from a randomized trial in sepsis. Thromb Haemost 2009;101:139-44.
24. Wahby KA, Riley LK, Tennenberg SD. Assessment of an extended interval fondaparinux dosing regimen for venous thromboembolism prophylaxis in critically ill patients with severe renal dysfunction using antifactor Xa levels. Pharmacotherapy 2017;37:1241-48.

**End-of-Life Care**

1. Brody H, Campbell ML, Faber-Langendoen K, et al. Withdrawing intensive life-sustaining treatment – recommendations for compassionate clinical management. N Engl J Med 1997;336:652-7.
2. Chan JD, Treece PD, Engelberg RA, et al. Narcotic and benzodiazepine use after withdrawal of life support: association with time to death? Chest 2004;126:286-93.
3. Global Atlas of Palliative Care at the End of Life. Available at https://www.who.int/nmh/Global_Atlas_of_Palliative_Care.pdf.
4. Kamal AJ, Maguire JM, Wheeler JL, et al. Dyspnea review for the palliative care professional: treatment goals and therapeutic options. J Palliat Med 2012;15:106-14.
5. Laserna A, Durán-Crane A, López-Olivo MA, et al. Pain management during the withholding and withdrawal of life support in critically ill patients at the end-of-life: a systematic review and meta-analysis. Intensive Care Med. 2020;46:1671-82.
6. Simon ST, Higginson IJ, Booth S, et al. Benzodiazepines for the relief of breathlessness in advanced malignant and non-malignant diseases in adults. Cochrane Database Syst Rev 2010;1:CD007354.
7. Treece PD, Engelberg RA, Crowley L, et al. Evaluation of a standardized order form for the withdrawal of life support in the intensive care unit. Crit Care Med 2004;32:1141-8.
8. Truog RD, Campbell ML, Curtis JR, et al. Recommendations for end-of-life care in the intensive care unit: a consensus statement by the American College of Critical Care Medicine. Crit Care Med 2008;36:953-63.

9. Wilson WC, Smedira NG, Fink C, et al. Ordering and administration of sedatives and analgesics during the withholding and withdrawal of life support from critically ill patients. JAMA 1992;267:949-53.

**Disaster Management**

1. Aruru M, Truong HA, Clark S. Pharmacy Emergency Preparedness and Response (PEPR): a proposed framework for expanding pharmacy professionals' roles and contributions to emergency preparedness and response during the COVID-19 pandemic and beyond. Res Social Adm Pharm 2021;17:1967-77.
2. CDC Hospital All-Hazards Self-Assessment. Available at https://www.cdc.gov/orr/readiness/resources/healthcare/documents/hah_508_compliant_final.pdf.
3. FEMA Glossary. Available at https://training.fema.gov/programs/emischool/el361toolkit/glossary.htm.
4. Raza MA, Aziz S, Noreen M, et al. Role of pharmacist in disaster management: a quantitative content analysis approach. Innov Pharm 2021;12:10.24926/iip.v12i4.4359.
5. Setlak P. Bioterrorism preparedness and response: emerging role of health-system pharmacists. Am J Health Syst Pharm 2004;61(11):1167-75.
6. Wax Rs. Preparing the Intensive Care Unit for Diaster. Crit Care Clin.2019 Oct;35(4):551-562.

**Social Determinants of Health**

1. Barwise B, Wi C, Frank R, et al. An Innovative Individual-Level Socioeconomic Measure Predicts Critical Care Outcomes in Older Adults: A Population-Based Study. J Intensive Care Med. 2021;36:828-37.
2. Bastian K, Hollinger A, Mebazaa A, et al. Association of social deprivation with 1-year outcome of ICU survivors: results from the FROG-ICU study. Intensive Care Med. 2018 Dec;44(12):2025-2037.
3. Celedón JC, Burchard EG, Schraufnagel D, et al. An American Thoracic Society/National Heart, Lung, and Blood Institute Workshop Report: Addressing Respiratory Health Equality in the United States. Ann Am Thorac Soc 2017;14(5):814-26.
4. Chertoff J. Racial disparities in critical care: experience from the USA. Lancet Respir Med. 2017 Feb;5(2):e11-e12.
5. Falvey JR, Murphy TE, Leo-Summers L, et al. Neighborhood Socioeconomic Disadvantage and Disability After Critical Illness. Crit Care Med 2022;50(5):733-41.
6. Goodwin AJ, Nadig NR, McElligott JT, et al. Where You Live Matters: The Impact of Place of Residence on Severe Sepsis Incidence and Mortality. Chest. 2016 Oct;150(4):829-836.
7. HHS, Office of Disease Prevention and Health Promotion, Healthy People. Available at https://health.gov/healthypeople.
8. HHS, Office of the Assistant Secretary for Planning and Evaluation. Building the Evidence Base for Social Determinants of Health Interventions. Available at https://aspe.hhs.gov/reports/building-evidence-base-social-determinants-health-interventions.
9. Jones JRA, Berney S, Connolly B, et al. Socioeconomic Position and Health Outcomes Following Critical Illness: A Systematic Review. Crit Care Med. 2019 Jun;47(6):e512-e521.
10. Lyon SM, Benson NM, Cooke CR, et al. The effect of insurance status on mortality and procedural use in critically ill patients. Am J Respir Crit Care Med 2011;184:809-15.
11. Potter KM, Scheunemann LP, Girard TD. Health Equity and Critical Care Survivorship: Where Do We Go From Here? Crit Care 2019;35:551-62.

## ANSWERS AND EXPLANATIONS TO PATIENT CASES

**1. Answer: A**

The mnemonic FAST-HUG stands for **F**eeding, **A**nalgesia, **S**edation, **T**hromboembolic prophylaxis, **H**ead of bed elevation, stress **U**lcer prophylaxis, and **G**lycemic control. Using this mnemonic as a checklist every day for each critically ill patient will help maximize therapeutic interventions and promote patient safety. This patient would benefit from receiving enteral nutrition (an NGT is already placed and he has a working GI tract), interrupting the sedative (current RASS score is above the designated goal), and adding SUP (risk factors include mechanical ventilation) (Answer A is correct). Critically ill patients with risk factors for VTE should remain on VTE prophylaxis (Answers B and D are incorrect); moreover, sliding-scale insulin should be initiated when the patient is not critically ill, adding another reason why Answer B is incorrect and making Answer C incorrect.

**2. Answer: C**

Two independent risk factors for SRMD are respiratory failure requiring mechanical ventilation for 48 hours or longer and coagulopathy (Plt less than 50,000/$mm^3$, INR greater than 1.5, or aPTT greater than 2 times the control). This patient has both of these risk factors. In addition, she has septic shock, as evidenced by end-organ dysfunction and acute kidney injury (Answer C is correct). Answers A, B, and D are incorrect because this patient has four risk factors for developing stress-related mucosal damage.

**3. Answer: D**

Antacids are not recommended for routine use because of their frequency of administration, adverse effects, and interactions (Answer B is incorrect). In a large randomized controlled trial, sucralfate was inferior to $H_2$RAs in preventing clinically significant bleeding from SRMD and is generally not recommended because of its adverse effect profile (Answer A is incorrect). Proton pump inhibitors are no better than $H_2$RAs in preventing SRMD and are associated with increased infectious complications, including pneumonia and CDI (Answer D is correct). Meta-analyses have favored PPIs to $H_2$RAs for GI bleeding; however, the individual trials included lacked methodological quality (Answer C is incorrect).

**4. Answer: C**

Once the risk factors are no longer present, SUP should promptly be discontinued (Answer C is correct). This patient no longer has risk factors (mechanical ventilation, coagulopathy acute kidney failure, and severe sepsis). In addition, there is no evidence that SUP should be continued until hospital or ICU discharge or when antimicrobial therapy is complete (Answers A, B, and D are incorrect).

**5. Answer: C**

This patient has several risk factors for VTE, including immobility and respiratory failure, making heparin 5000 units subcutaneously twice daily appropriate for VTE prophylaxis (Answer C is correct). Neither enoxaparin nor fondaparinux is appropriate for this patient, who has acute kidney injury with an estimated CrCl of less than 20 mL/minute/1.73 $m^2$ (Answers B and D are incorrect). Additionally, VTE prophylaxis with fondaparinux is contraindicated in patients less than 50 kg. Intermittent pneumatic compression would be insufficient in a patient with no contraindication to pharmacologic prophylaxis (Answer A is incorrect).

**6. Answer: A**

Diagnosing HIT is difficult in a critically ill patient because there are many alternative causes of thrombocytopenia. Clinical assessment is essential in diagnosing HIT because of the immediate need for treatment and the delay in laboratory testing (Answers B and D are incorrect). Clinically, this patient has had a greater than 50% decrease in Plt within 5 days of receiving heparin. The calculated 4Ts score is 5 (2 points for Plt decrease by greater than 50%, 2 points for a clear onset at days 5–10, and 1 point for other possible causes of thrombocytopenia. This score is an intermediate probability for HIT. In managing suspected HIT, first ensure that all forms of heparin are discontinued, including flushes and heparin-coated catheters. Next, initiate an alternative form of anticoagulation. Direct thrombin inhibitors are the agents of choice for anticoagulation in acute HIT because they have no cross-reactivity with heparin (Answer A is correct). Factor Xa inhibitors have been used to manage HIT; however, they would not be best in this patient, who has acute kidney injury (Answer C is incorrect).

**7. Answer: D**

Warfarin can be initiated (Answer C is incorrect) once the Plt has recovered to at least 150,000/mm$^3$ and after at least 5 days of therapy with an alternative anticoagulant (Answer D is correct). Because this patient's Plts have not reached 150,000/mm$^3$ and only 3 days of argatroban have been completed, warfarin therapy should not be initiated at this time (Answers A and B are incorrect). Argatroban should be continued, and warfarin may be considered at low doses (maximum 5 mg) as the Plt continues to recover (Answer B is incorrect).

**8. Answer: D**

Up to 50% of seriously ill hospitalized patients have moderate or severe pain. Opioids are the treatment mainstay for patients with pain and dyspnea at the end of life. Assessing pain in the ICU can be particularly challenging because many patients have impaired cognition and communication. Vital signs alone should not be used for pain assessment (Answer A is incorrect). Evidence suggests that pain can be improved with correct dosing and titration (Answers B and C are incorrect) without causing respiratory depression or hastening death (Answer D is correct).

## ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

**1. Answer: C**

The FAST-HUG mnemonic can serve as a checklist for every patient admitted to the ICU. Every patient should be assessed for a sedation interruption to minimize sedative exposure and maintain a light level of sedation (Answer C is correct). To decrease the risk of nosocomial pneumonia, each patient should have his or her head elevated 30–45 degrees above the head of the bed (Answer C is correct). Enteral nutrition should be initiated as soon as possible – typically, once the patient is stabilized; however, thromboprophylaxis should be initiated in every patient, using pharmacologic agents preferentially to mechanical prophylaxis (Answer B is incorrect). Stress ulcer prophylaxis should only be initiated in patients who have risk factors and should be discontinued once the risk factors no longer exist (Answer A is incorrect). Insulin infusions should be initiated only if blood glucose readings are not 140–180 mg/dL (Answer D is incorrect).

**2. Answer: C**

Sucralfate forms a protective barrier over the surface of the stomach, reducing exposure to acidic gastric contents; therefore, sucralfate does not affect gastric pH (Answer A is incorrect). Compared with $H_2$RAs, PPIs appear more effective at reducing gastric acidity, but no well-conducted trial has shown PPIs superior in preventing clinically significant bleeding (Answer B is incorrect). Tolerance to any $H_2$RA may occur, but not to PPIs (Answer C is correct). Antacids have some effect on reducing stress ulceration, provided the gastric pH is kept around 3.5, but frequent dosing (up to every 2 hours) is required to achieve this goal, making their use impractical (Answer D is incorrect).

**3. Answer: D**

The patient has an indication for SUP (mechanical ventilation). He has an NGT in place and is tolerating tube feedings, indicating a functioning gut; therefore, intravenous therapy is not required (Answers A and B are incorrect). The patient has erosive esophagitis, for which a PPI will be more effective than an $H_2$RA (Answer C is incorrect). Omeprazole suspension is effective in preventing SRMD; therefore, an omeprazole suspension would be most appropriate for this patient (Answer D is correct).

**4. Answer: A**

Proton pump inhibitors are potent inhibitors of gastric acid production and are the drug of choice for gastroesophageal reflux disease. To date, the only prospective randomized controlled trial to evaluate CDI risk with PPI use found no difference between PPI and placebo; however, this was an underpowered secondary end point, and several cohort studies have found an association. (Answer C is incorrect). All published trials assessing the risk of CDI with PPI use have been limited by the inconsistent definitions of CDI and the variable infection control practices (Answer D is incorrect). Gastric juice is strongly bactericidal for microorganisms. Proton pump inhibitors are commonly used to increase the gastric pH; therefore, they act as a potential risk factor for CDI (Answer A is correct).

**5. Answer: B**

Low-dose unfractionated heparin or low-molecular-weight heparin should be initiated for VTE prophylaxis in a critically ill patient over no prophylaxis (Answer D is incorrect). Intermittent pneumatic compression devices would be insufficient prophylaxis in a patient with several risk factors for VTE (Answer A is incorrect). A continuous infusion of heparin is inappropriate for preventing VTE (Answer C is incorrect). Enoxaparin may be used for VTE prophylaxis in a critically ill patient with stable renal function (Answer B is correct).

**6. Answer: A**

This patient had a closed-head injury, placing her at high risk of VTE (Answer D is incorrect). She is at high risk of major bleeding and acute kidney injury; therefore, a low-molecular-weight heparin or a factor Xa inhibitor would not be best for her (Answers B and C are incorrect). Mechanical prophylaxis with intermittent pneumatic compression devices is preferred to no prophylaxis in the absence of lower-extremity injury until the bleeding risk is no longer present (Answer A is correct).

**7. Answer: D**

Clinical assessment is essential in diagnosing HIT because of the immediate need for treatment and the delay in laboratory testing. Although this patient's Plt did decrease by 50%, the characteristic onset of the Plt

decrease in HIT is 5–10 days after heparin initiation. Clinical prediction rules to help determine the probability of HIT (e.g., 4Ts score) have been developed. Patients with a low 4Ts score (0–3) have a very low probability of HIT (Answer D is correct). Direct thrombin inhibitors are the agents of choice for anticoagulation in acute HIT because they have no cross-reactivity with heparin. Initiating these agents in those with a low probability of HIT could lead to an unnecessary increase in bleeding risk (Answer B is incorrect). If the 4T score revealed an intermediate or high risk of HIT, the first step would be to ensure that all forms of heparin are discontinued, including flushes and heparin-coated catheters (Answer C is incorrect). The next step would be to initiate an alternative form of anticoagulation (Answer A is incorrect).

**8. Answer: C**

General considerations in the critically ill patient at the end of life include minimizing uncomfortable or unnecessary procedures, tests, and treatments, including fingersticks, Foley catheters, and routine vital signs (Answers A, B, and D are incorrect). Symptom management of pain and anxiety, fever, cough, secretions, nausea and vomiting, and delirium should be considered in the dying patient (Answer C is correct).